

StormSense AI IMD/CWC Operational Documentation Dossier

Version: 2.1

Generated: 2026-06-08

Prepared for: IMD/CWC Review, Academic Submission, Internship Evaluation, Hackathon Presentation, Production Handover, Technical Review, and Deployment Documentation.

Document Control

Field	Value
Document Type	IMD/CWC operational review and production handover dossier
Evidence Basis	Supplied UI screenshots, frontend source code, backend source code, Supabase schema, generated migration/deployment reports, and historical archive files
Source Repository	C:\Users\USER\Thunderstorm analysis
Screenshot Evidence	46 supplied screenshots copied into `docs/screenshots/`
Prepared Date	2026-06-08

Version History

Version	Date	Summary
1.0	Phase 2/3	Deterministic convective forecasting, archive, verification, and operations modules.
1.5	Phase 5.9	Reviewer polish, file analysis center, registry, typography, and deployment refinement.
2.0	2026-06-08	Supabase migration, single-URL Render deployment, and complete IMD/CWC documentation dossier.
2.1	2026-06-08	Screenshot-backed evidence catalogue, standalone audit deliverables, and front-matter simplified for reviewer use.

Acknowledgement

StormSense AI was developed as an operational atmospheric intelligence workstation focused on thunderstorm forecasting, verification, historical analysis, and IMD/CWC-style review workflows. The system integrates live/cached sounding ingestion, deterministic forecast engines, historical archives, verification science, and production deployment configuration.

Executive Summary

StormSense AI is a FastAPI + Vite React atmospheric decision-support platform. Repository inspection found 91 source/data/runtime files under `backend/` and `frontend/src/`, 54 FastAPI route decorators, 12 Supabase schema tables, 360 RSRW historical archive records, and 240 IMD observational records. The application provides radar-like monitoring, deterministic forecast generation, historical thunderstorm archive workflows, file upload analysis, verification metrics, district impact intelligence, operational bulletins, review mode, and single-URL Render deployment.

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1. Project Overview

Introduction

StormSense AI is an IMD-style atmospheric intelligence platform designed to support severe weather monitoring and thunderstorm forecasting decisions. It combines operational frontend workstations with backend forecast, verification, archive, and research engines.

Project Objectives

- Interpret sounding and historical weather datasets in operational terms.
- Produce deterministic thunderstorm forecasts and probability diagnostics.
- Verify forecasts scientifically using CSI, POD, FAR, HSS, and BIAS.
- Support IMD/CWC review through simplified review mode and reviewer dashboard.
- Preserve historical archive workflows for 2023, 2024, and 2025 records.
- Serve frontend and backend from one Render URL for production readiness.

Problem Statement

Thunderstorm forecasting review requires a joined workflow: sounding interpretation, convective index calculation, forecast reasoning, observed-event verification, district impact assessment, and reviewer decision capture. Spreadsheet-only or display-only systems make it difficult to trace why a forecast was issued, whether it verified, and how thresholds should be tuned.

Need for Thunderstorm Forecast Verification

Operational confidence depends on observed-event validation. StormSense AI provides contingency analysis, analog comparison, threshold reliability, archive case review, and probability traceability so that forecast behavior can be reviewed scientifically rather than treated as opaque dashboard output.

IMD/CWC Operational Relevance

The system emphasizes Visakhapatnam, Machilipatnam, north coastal Andhra, radiosonde cycle metadata, district-level impact narratives, lightning/heavy rain/squall guidance, and operational bulletin workflows. It uses station codes, synoptic cycles, deterministic thresholds, and IMD-style terms throughout the review experience.

Scope of StormSense AI

The platform covers live/cached sounding ingestion, forecast generation, radar-sounding-map fusion, historical archive analysis, file upload analysis, probabilistic forecasting, verification science, climatology, analog intelligence, reviewer dashboards, and single-service deployment.

Key Capabilities

Capability	Implementation Evidence
Live/cached sounding ingestion	`backend/fetch_sounding.py`, `/cwc/sounding-raw/{station_code}`, CAPE traceability metadata
Forecast engine	`backend/main.py`, `fetch_station_data`, `/forecast`, `PredictorEngine.jsx`
Verification science	`backend/analysis_engines.py`, `/cwc/verification`, `/cwc/threshold-research`, `ResearchHub.jsx`
Historical archive	`backend/data/rsnw_historical_archive.*`, `/cwc/historical-dates`, `ResearchHub.jsx:HISTORICAL_WORKBENCH`
File analysis center	`/cwc/analyze-historical-dataset`, `ResearchHub.jsx`, `Phase3OpsModule.jsx`
District impact	`compute_district_impact`, `/cwc/district-impact`, Andhra monitoring module

Capability	Implementation Evidence
Single URL deployment	`render.yaml`, static serving in `backend/main.py`

2. System Architecture

Architecture Diagrams

Generated from repository inspection on 2026-06-08.

High-Level Architecture

```
flowchart LR
    Operator["IMD/CWC Reviewer or Forecaster"] --> UI["Vite React Workstation"]
    UI --> API["FastAPI Operational Backend"]
    API --> Sounding["Wyoming / Cached Sounding Ingestion"]
    API --> Engines["Thermodynamic, Forecast, Verification, Research Engines"]
    API --> DB["Supabase PostgreSQL"]
    API --> Files["CSV/XLS/XLSX Upload and Export Workflows"]
    Engines --> Outputs["Nowcast, Verification, Registry, Bulletins, District Impacts"]
```

Component Architecture

```
flowchart TB
    App["frontend/src/App.jsx"] --> Sidebar["Sidebar Navigation"]
    App --> Radar["RadarConsole"]
    App --> Predictor["PredictorEngine"]
    App --> Analytics["AnalyticsDeck"]
    App --> Research["ResearchHub"]
    App --> Phase3["Phase3OpsModule"]
    App --> Health["HealthConsole"]
    Research --> Archive["Historical Thunderstorm Archive"]
    Research --> Simulator["Forecast Simulator"]
    Research --> Verify["Forecast Verification"]
    Phase3 --> Andhra["Coastal Thunderstorm Monitoring Center"]
    Phase3 --> Bulletin["Auto IMD Bulletin Generator"]
```

Deployment Architecture

```
flowchart LR
    Render["Single Render Web Service"] --> Build["Build: pip install + npm install + npm run build"]
    Build --> Dist["frontend/dist"]
    Render --> Uvicorn["uvicorn backend.main:app"]
    Uvicorn --> Static["FastAPI serves /assets and SPA fallback"]
    Uvicorn --> API["FastAPI API routes"]
    API --> Supabase["Supabase PostgreSQL"]
```

Authentication Flow

```
sequenceDiagram
    participant User
    participant React
    participant FastAPI
    participant DB as Supabase users table
    User->>React: Enter login credentials
    React->>FastAPI: POST /auth/login
    FastAPI->>DB: Lookup user by email
    DB-->>FastAPI: User record
    FastAPI-->>React: Bearer token and profile
    React->>FastAPI: Authenticated requests with Authorization header
    FastAPI-->>React: Protected operational outputs
```

Forecast Pipeline

```
flowchart TD
    Cycle["00Z / 12Z Cycle Lock"] --> Sounding["Sounding Fetch and Cache"]
    Sounding --> Thermo["Thermodynamic Solver"]
    Thermo --> Indices["CAPE, CIN, LI, PWAT, SWEAT, K-index"]
    Indices --> Prob["Probabilistic Forecast Engine"]
    Indices --> Lifecycle["Convective Lifecycle Engine"]
    Prob --> Decision["Operational Decision Support"]
    Lifecycle --> Decision
    Decision --> Bulletin["Bulletin / Watch / Advisory Outputs"]
    Decision --> Verification["Verification and Trust Scoring"]
```

Frontend Architecture

Primary navigation is implemented in `Sidebar.jsx` with 15 command deck entries. `App.jsx` manages authentication, review mode, forecast/historical/trend/escalation state, websocket connection, and module routing.

Navigation ID	Visible Label
RADAR	Live Doppler Radar
PREDICTOR	AI Prediction Engine
ANALYTICS	Convective Analytics
RESEARCH	Research & Insights
HEALTH	Health & Migrations
ATM_INTEL	Atmospheric Intelligence Center
WATCHDESK	Severe Weather Watchdesk
THERMO_LAB	Thermodynamic Lab
CLIMO_RESEARCH	Climatology & Research Center
AI_LAB	Thunderstorm Forecast Simulator
RADSAT_FUSION	Live Operational Nowcast Center
VERIFY_CENTER	Forecast Verification
ANALOG_ARCHIVE	Historical Thunderstorm Archive
BULLETIN	Auto IMD Bulletin Generator
ANDHRA_MONITOR	Coastal Thunderstorm Monitoring Center

Backend Architecture

The FastAPI backend is centered in `backend/main.py`. It owns authentication routes, forecast routes, websocket stream, system status, CWC/research endpoints, file upload endpoints, and static frontend serving. Scientific logic is delegated to `analysis_engines.py`, `thermo.py`, and `fetch_sounding.py`.

Engine Architecture

Engine	Source	Role
Thermodynamic solver	backend/thermo.py	Sounding parsing, CAPE/CIN/LCL/LFC/EL and parcel-path calculations.
Sounding ingestion	backend/fetch_sounding.py	Wyoming fetch, cache metadata, cycle-aware source status, freshness scoring.
Research and verification	backend/analysis_engines.py	Historical observations, contingency metrics, threshold research, probability, climatology, analogs, district impact.
Forecast orchestration	backend/main.py	Station forecast assembly, lifecycle, decision support, persistence, websocket output.
ML training artifact	backend/ml/train_model.py and storm_model.pkl	Random forest training script and model artifact for storm classification support.

Page-by-Page Documentation

Purpose	Primary Users	Inputs
Authenticated landing workspace that coordinates HUD status, station forecasts, websocket telemetry, and module routing.	Forecaster or reviewer.	Login token, forecast API payloads, websocket
Radar-like operational visualization driven by forecast data, lifecycle state, lightning, rainfall, and station risk.	Nowcasting forecaster.	Forecast rows, storm probability, station metac
Station-level deterministic forecast workstation with convective index interpretation.	Forecast desk reviewer.	CAPE, LI, SWEAT, PWAT, K-index, cycle data

	Purpose	Primary Users	Inputs
	Verification, analytics, climatology, and decision-support panels.	Research analyst and verification reviewer.	Historical observations, thresholds, forecast ro
	Multi-tab research center containing archive, simulator, verification, database, review, glossary, and onboarding pages.	Reviewer, researcher, and presenter.	Historical archive, upload file, selected station/
NCH	Flagship meteorological investigation console for historical thunderstorm records.	IMD/CWC reviewer.	2023-2025 archive records, file upload, station.
	Historical and custom sounding forecast reproduction lab.	Forecaster and academic evaluator.	Station, date, custom sounding values, upload
	Threshold testing and contingency-matrix verification workspace.	Verification scientist.	Thresholds, station, season, historical records.
	Historical weather database browser.	Research analyst.	Archive index and selected station/date filters.
D	Reviewer docket and operational review console.	Reviewer or evaluator.	Selected event, reviewer name, docket ID, con
	Operational reference page for purpose, workflow, data sources, architecture, and outputs.	Reviewer and evaluator.	Static project metadata and workflow descripti
	Reviewer onboarding center.	First-time reviewer.	None beyond app state.
xx	Upload and audit historical datasets.	Research analyst and reviewer.	CSV, XLS, XLSX, radiosonde, and sounding d
	Severe watchdesk, Andhra monitoring, radar-sounding fusion, and operational intelligence modules.	Nowcasting desk.	Forecast data, lifecycle state, station focus, dis
	Operational bulletin generation workspace.	Forecaster preparing advisory text.	Current forecast rows and cycle metadata.
	Deployment/runtime health and migration visibility console.	Developer/operator.	API base URL, websocket URL, system status

Screenshot evidence note: 46 supplied UI screenshots are copied under `docs/screenshots/` and catalogued in `SCREENSHOT\_ANALYSIS\_REPORT.md`.

Dashboard / Operational Shell

Source reference: `frontend/src/App.jsx`.

Purpose: Authenticated landing workspace that coordinates HUD status, station forecasts, websocket telemetry, and module routing.

Who uses it: Forecaster or reviewer.

Inputs: Login token, forecast API payloads, websocket stream, cycle info.

Outputs: Operational module views, HUD status, active cycle, and navigation context.

Workflow: open the page from the sidebar or Research & Insights tab, confirm the operational metadata HUD, select station/date/cycle or file where applicable, review the current situation, meteorological interpretation, and recommended action, then export or record the reviewer verdict when the workflow supports it.

Operational relevance: this page contributes to the decision chain by answering what is happening, why it is happening, what may happen next, and what operational action should be taken.

Example usage: during IMD/CWC review, demonstrate the page with Visakhapatnam and Machilipatnam focus where station selection is available, then connect the visible output to the relevant backend endpoint or archive record.

Screenshot reference: see the screenshot evidence catalogue and `SCREENSHOT\_ANALYSIS\_REPORT.md` for matching supplied UI evidence.

Live Doppler Radar

Source reference: `frontend/src/components/modules/RadarConsole.jsx`.

Purpose: Radar-like operational visualization driven by forecast data, lifecycle state, lightning, rainfall, and station risk.

Who uses it: Nowcasting forecaster.

Inputs: Forecast rows, storm probability, station metadata.

Outputs: Operational radar console, severe markers, and monitoring interpretation.

Workflow: open the page from the sidebar or Research & Insights tab, confirm the operational metadata HUD, select station/date/cycle or file where applicable, review the current situation, meteorological interpretation, and recommended action, then export or record the reviewer verdict when the workflow supports it.

Operational relevance: this page contributes to the decision chain by answering what is happening, why it is happening, what may happen next, and what operational action should be taken.

Example usage: during IMD/CWC review, demonstrate the page with Visakhapatnam and Machilipatnam focus where station selection is available, then connect the visible output to the relevant backend endpoint or archive record.

Screenshot reference: see the screenshot evidence catalogue and `SCREENSHOT\_ANALYSIS\_REPORT.md` for matching supplied UI evidence.

AI Prediction Engine

Source reference: `frontend/src/components/modules/PredictorEngine.jsx`.

Purpose: Station-level deterministic forecast workstation with convective index interpretation.

Who uses it: Forecast desk reviewer.

Inputs: CAPE, LI, SWEAT, PWAT, K-index, cycle data.

Outputs: Forecast classification, probability, trend, and action guidance.

Workflow: open the page from the sidebar or Research & Insights tab, confirm the operational metadata HUD, select station/date/cycle or file where applicable, review the current situation, meteorological interpretation, and recommended action, then export or record the reviewer verdict when the workflow supports it.

Operational relevance: this page contributes to the decision chain by answering what is happening, why it is happening, what may happen next, and what operational action should be taken.

Example usage: during IMD/CWC review, demonstrate the page with Visakhapatnam and Machilipatnam focus where station selection is available, then connect the visible output to the relevant backend endpoint or archive record.

Screenshot reference: see the screenshot evidence catalogue and `SCREENSHOT\_ANALYSIS\_REPORT.md` for matching supplied UI evidence.

Convective Analytics

Source reference: `frontend/src/components/modules/AnalyticsDeck.jsx`.

Purpose: Verification, analytics, climatology, and decision-support panels.

Who uses it: Research analyst and verification reviewer.

Inputs: Historical observations, thresholds, forecast rows.

Outputs: Skill metrics, reliability, threshold insights, and forecast trust context.

Workflow: open the page from the sidebar or Research & Insights tab, confirm the operational metadata HUD, select station/date/cycle or file where applicable, review the current situation, meteorological interpretation, and recommended action, then export or record the reviewer verdict when the workflow supports it.

Operational relevance: this page contributes to the decision chain by answering what is happening, why it is happening, what may happen next, and what operational action should be taken.

Example usage: during IMD/CWC review, demonstrate the page with Visakhapatnam and Machilipatnam focus where station selection is available, then connect the visible output to the relevant backend endpoint or archive record.

Screenshot reference: see the screenshot evidence catalogue and `SCREENSHOT\_ANALYSIS\_REPORT.md` for matching supplied UI evidence.

Research & Insights

Source reference: `frontend/src/components/modules/ResearchHub.jsx`.

Purpose: Multi-tab research center containing archive, simulator, verification, database, review, glossary, and onboarding pages.

Who uses it: Reviewer, researcher, and presenter.

Inputs: Historical archive, upload file, selected station/date, thresholds.

Outputs: Case analysis, registry, verification, dataset audit, and reviewer docket.

Workflow: open the page from the sidebar or Research & Insights tab, confirm the operational metadata HUD, select station/date/cycle or file where applicable, review the current situation, meteorological interpretation, and recommended action, then export or record the reviewer verdict when the workflow supports it.

Operational relevance: this page contributes to the decision chain by answering what is happening, why it is happening, what may happen next, and what operational action should be taken.

Example usage: during IMD/CWC review, demonstrate the page with Visakhapatnam and Machilipatnam focus where station selection is available, then connect the visible output to the relevant backend endpoint or archive record.

Screenshot reference: see the screenshot evidence catalogue and `SCREENSHOT\_ANALYSIS\_REPORT.md` for matching supplied UI evidence.

Historical Thunderstorm Archive

Source reference: `frontend/src/components/modules/ResearchHub.jsx:HISTORICAL\_WORKBENCH`.

Purpose: Flagship meteorological investigation console for historical thunderstorm records.

Who uses it: IMD/CWC reviewer.

Inputs: 2023-2025 archive records, file upload, station/date filters.

Outputs: Latest records, thunderstorm registry, archive summaries, exports, and historical analysis.

Workflow: open the page from the sidebar or Research & Insights tab, confirm the operational metadata HUD, select station/date/cycle or file where applicable, review the current situation, meteorological interpretation, and recommended action, then export or record the reviewer verdict when the workflow supports it.

Operational relevance: this page contributes to the decision chain by answering what is happening, why it is happening, what may happen next, and what operational action should be taken.

Example usage: during IMD/CWC review, demonstrate the page with Visakhapatnam and Machilipatnam focus where station selection is available, then connect the visible output to the relevant backend endpoint or archive record.

Screenshot reference: see the screenshot evidence catalogue and `SCREENSHOT\_ANALYSIS\_REPORT.md` for matching supplied UI evidence.

Thunderstorm Forecast Simulator

Source reference: `frontend/src/components/modules/ResearchHub.jsx:FORECAST\_LAB`.

Purpose: Historical and custom sounding forecast reproduction lab.

Who uses it: Forecaster and academic evaluator.

Inputs: Station, date, custom sounding values, uploaded dataset.

Outputs: Simulated forecast, threshold trace, probability, verification, and exportable report.

Workflow: open the page from the sidebar or Research & Insights tab, confirm the operational metadata HUD, select station/date/cycle or file where applicable, review the current situation, meteorological interpretation, and recommended action, then export or record the reviewer verdict when the workflow supports it.

Operational relevance: this page contributes to the decision chain by answering what is happening, why it is happening, what may happen next, and what operational action should be taken.

Example usage: during IMD/CWC review, demonstrate the page with Visakhapatnam and Machilipatnam focus where station selection is available, then connect the visible output to the relevant backend endpoint or archive record.

Screenshot reference: see the screenshot evidence catalogue and `SCREENSHOT\_ANALYSIS\_REPORT.md` for matching supplied UI evidence.

Forecast Verification

Source reference: `frontend/src/components/modules/ResearchHub.jsx:RESEARCH\_VERIFY`.

Purpose: Threshold testing and contingency-matrix verification workspace.

Who uses it: Verification scientist.

Inputs: Thresholds, station, season, historical records.

Outputs: CSI, POD, FAR, HSS, BIAS, and recommended calibration direction.

Workflow: open the page from the sidebar or Research & Insights tab, confirm the operational metadata HUD, select station/date/cycle or file where applicable, review the current situation, meteorological interpretation, and recommended action, then export or record the reviewer verdict when the workflow supports it.

Operational relevance: this page contributes to the decision chain by answering what is happening, why it is happening, what may happen next, and what operational action should be taken.

Example usage: during IMD/CWC review, demonstrate the page with Visakhapatnam and Machilipatnam focus where station selection is available, then connect the visible output to the relevant backend endpoint or archive record.

Screenshot reference: see the screenshot evidence catalogue and `SCREENSHOT\_ANALYSIS\_REPORT.md` for matching supplied UI evidence.

Dataset Explorer

Source reference: `frontend/src/components/modules/ResearchHub.jsx:DATASET\_EXPLORER`.

Purpose: Historical weather database browser.

Who uses it: Research analyst.

Inputs: Archive index and selected station/date filters.

Outputs: Weather archive log and searchable historical observations.

Workflow: open the page from the sidebar or Research & Insights tab, confirm the operational metadata HUD, select station/date/cycle or file where applicable, review the current situation, meteorological interpretation, and recommended action, then export or record the reviewer verdict when the workflow supports it.

Operational relevance: this page contributes to the decision chain by answering what is happening, why it is happening, what may happen next, and what operational action should be taken.

Example usage: during IMD/CWC review, demonstrate the page with Visakhapatnam and Machilipatnam focus where station selection is available, then connect the visible output to the relevant backend endpoint or archive record.

Screenshot reference: see the screenshot evidence catalogue and `SCREENSHOT\_ANALYSIS\_REPORT.md` for matching supplied UI evidence.

IMD Review Dashboard

Source reference: `frontend/src/components/modules/ResearchHub.jsx:REVIEWER\_DASHBOARD`.

Purpose: Reviewer docket and operational review console.

Who uses it: Reviewer or evaluator.

Inputs: Selected event, reviewer name, docket ID, comments.

Outputs: Review summary, verdict, action, export metadata.

Workflow: open the page from the sidebar or Research & Insights tab, confirm the operational metadata HUD, select station/date/cycle or file where applicable, review the current situation, meteorological interpretation, and recommended action, then export or record the reviewer verdict when the workflow supports it.

Operational relevance: this page contributes to the decision chain by answering what is happening, why it is happening, what may happen next, and what operational action should be taken.

Example usage: during IMD/CWC review, demonstrate the page with Visakhapatnam and Machilipatnam focus where station selection is available, then connect the visible output to the relevant backend endpoint or archive record.

Screenshot reference: see the screenshot evidence catalogue and `SCREENSHOT\_ANALYSIS\_REPORT.md` for matching supplied UI evidence.

About StormSense AI

Source reference: `frontend/src/components/modules/ResearchHub.jsx:ABOUT`.

Purpose: Operational reference page for purpose, workflow, data sources, architecture, and outputs.

Who uses it: Reviewer and evaluator.

Inputs: Static project metadata and workflow descriptions.

Outputs: Architecture and operational purpose narrative.

Workflow: open the page from the sidebar or Research & Insights tab, confirm the operational metadata HUD, select station/date/cycle or file where applicable, review the current situation, meteorological interpretation, and recommended action, then export or record the reviewer verdict when the workflow supports it.

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Screenshot reference: see the screenshot evidence catalogue and `SCREENSHOT\_ANALYSIS\_REPORT.md` for matching supplied UI evidence.

Start Here

Source reference: `frontend/src/components/modules/ResearchHub.jsx:START\_HERE`.

Purpose: Reviewer onboarding center.

Who uses it: First-time reviewer.

Inputs: None beyond app state.

Outputs: Data requirements, outputs, common review flow, and quick navigation.

Workflow: open the page from the sidebar or Research & Insights tab, confirm the operational metadata HUD, select station/date/cycle or file where applicable, review the current situation, meteorological interpretation, and recommended action, then export or record the reviewer verdict when the workflow supports it.

Operational relevance: this page contributes to the decision chain by answering what is happening, why it is happening, what may happen next, and what operational action should be taken.

Example usage: during IMD/CWC review, demonstrate the page with Visakhapatnam and Machilipatnam focus where station selection is available, then connect the visible output to the relevant backend endpoint or archive record.

Screenshot reference: see the screenshot evidence catalogue and `SCREENSHOT\_ANALYSIS\_REPORT.md` for matching supplied UI evidence.

File Analysis Center

Source reference: `frontend/src/components/modules/ResearchHub.jsx` and `Phase3OpsModule.jsx`.

Purpose: Upload and audit historical datasets.

Who uses it: Research analyst and reviewer.

Inputs: CSV, XLS, XLSX, radiosonde, and sounding datasets.

Outputs: Quality summary, column detection, registry, verification, and report exports.

Workflow: open the page from the sidebar or Research & Insights tab, confirm the operational metadata HUD, select station/date/cycle or file where applicable, review the current situation, meteorological interpretation, and recommended action, then export or record the reviewer verdict when the workflow supports it.

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Example usage: during IMD/CWC review, demonstrate the page with Visakhapatnam and Machilipatnam focus where station selection is available, then connect the visible output to the relevant backend endpoint or archive record.

Screenshot reference: see the screenshot evidence catalogue and `SCREENSHOT\_ANALYSIS\_REPORT.md` for matching supplied UI evidence.

Operational Monitoring

Source reference: `frontend/src/components/modules/Phase3OpsModule.jsx`.

Purpose: Severe watchdesk, Andhra monitoring, radar-sounding fusion, and operational intelligence modules.

Who uses it: Nowcasting desk.

Inputs: Forecast data, lifecycle state, station focus, district selection.

Outputs: Monitoring recommendation, district impacts, and severe-weather guidance.

Workflow: open the page from the sidebar or Research & Insights tab, confirm the operational metadata HUD, select station/date/cycle or file where applicable, review the current situation, meteorological interpretation, and recommended action, then export or record the reviewer verdict when the workflow supports it.

Operational relevance: this page contributes to the decision chain by answering what is happening, why it is happening, what may happen next, and what operational action should be taken.

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Screenshot reference: see the screenshot evidence catalogue and `SCREENSHOT\_ANALYSIS\_REPORT.md` for matching supplied UI evidence.

Bulletin Generator

Source reference: `frontend/src/components/modules/Phase3OpsModule.jsx:BULLETIN`.

Purpose: Operational bulletin generation workspace.

Who uses it: Forecaster preparing advisory text.

Inputs: Current forecast rows and cycle metadata.

Outputs: Thunderstorm nowcasts, lightning advisories, heavy rainfall bulletins, and district summaries.

Workflow: open the page from the sidebar or Research & Insights tab, confirm the operational metadata HUD, select station/date/cycle or file where applicable, review the current situation, meteorological interpretation, and recommended action, then export or record the reviewer verdict when the workflow supports it.

Operational relevance: this page contributes to the decision chain by answering what is happening, why it is happening, what may happen next, and what operational action should be taken.

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Screenshot reference: see the screenshot evidence catalogue and `SCREENSHOT\_ANALYSIS\_REPORT.md` for matching supplied UI evidence.

Health & Migrations

Source reference: `frontend/src/components/modules/HealthConsole.jsx`.

Purpose: Deployment/runtime health and migration visibility console.

Who uses it: Developer/operator.

Inputs: API base URL, websocket URL, system status.

Outputs: Backend health, connectivity, and deployment readiness status.

Workflow: open the page from the sidebar or Research & Insights tab, confirm the operational metadata HUD, select station/date/cycle or file where applicable, review the current situation, meteorological interpretation, and recommended action, then export or record the reviewer verdict when the workflow supports it.

Operational relevance: this page contributes to the decision chain by answering what is happening, why it is happening, what may happen next, and what operational action should be taken.

Example usage: during IMD/CWC review, demonstrate the page with Visakhapatnam and Machilipatnam focus where station selection is available, then connect the visible output to the relevant backend endpoint or archive record.

Screenshot reference: see the screenshot evidence catalogue and `SCREENSHOT\_ANALYSIS\_REPORT.md` for matching supplied UI evidence.

4. Screenshot Evidence Catalogue

Screenshot Evidence Catalogue

Supplied screenshot evidence files copied into `docs/screenshots/`: 46.

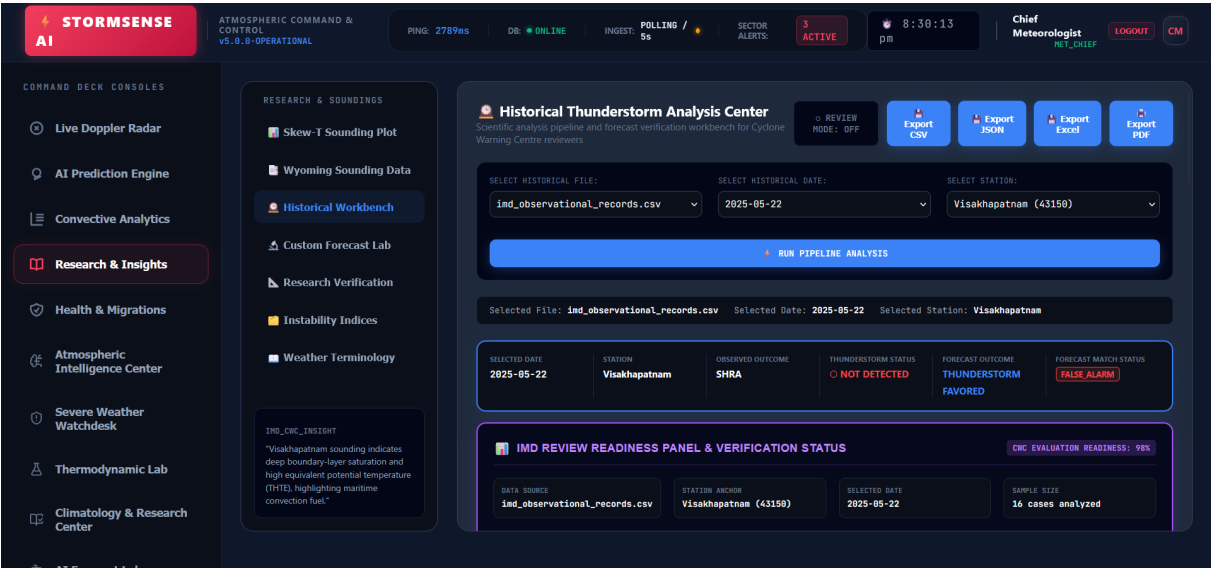
ID	UI Area	Evidence Summary	Local File
IMG-01	Historical Workbench	Historical thunderstorm analysis center with file/date/station selectors and IMD review readiness panel.	docs/screenshots/imd_evidence_01.png
IMG-02	Historical Workbench	Review readiness and verification status panel with contingency metrics and analog confidence.	docs/screenshots/imd_evidence_02.png
IMG-03	Historical Workbench	Pipeline evidence trail showing data ingestion and sounding parsing sections.	docs/screenshots/imd_evidence_03.png
IMG-04	Historical Workbench	Index calculation cards for CAPE, CIN, LI, PWAT, SWEAT, K-index, shear, and theta-e.	docs/screenshots/imd_evidence_04.png
IMG-05	Historical Workbench	Collapsed workflow stages for threshold comparison, verification, interpretation, and recommendation.	docs/screenshots/imd_evidence_05.png
IMG-06	Forecast Simulator	Custom convective sounding forecast lab with thermodynamic slider inputs and simulated forecast outcome.	docs/screenshots/imd_evidence_06.png
IMG-07	Forecast Verification	Convective index threshold verification lab with threshold controls and contingency metrics.	docs/screenshots/imd_evidence_07.png
IMG-08	AI Prediction Engine	Probabilistic forecast envelope and operational analog match evidence.	docs/screenshots/imd_evidence_08.png
IMG-09	AI Prediction Engine	Operational meteorologist decision support and composite convective severity score.	docs/screenshots/imd_evidence_09.png
IMG-10	Climatology & Research Center	Climatology and skill intelligence station view with seasonal recurrence composites.	docs/screenshots/imd_evidence_10.png
IMG-11	Climatology & Research Center	Expanded climatology view showing station composite metrics and chart area.	docs/screenshots/imd_evidence_11.png
IMG-12	Coastal Monitoring	Bay of Bengal coastal radar map focused on north coastal Andhra and station comparison.	docs/screenshots/imd_evidence_12.png
IMG-13	Coastal Monitoring	Coastal Andhra intelligence stream with marine inflow and corridor metrics.	docs/screenshots/imd_evidence_13.png
IMG-14	Research Navigation	Research & Soundings navigation stack with Dataset Explorer and Reviewer Audit Dashboard.	docs/screenshots/imd_evidence_14.png
IMG-15	Dataset Explorer	IMD Convective Dataset Explorer distribution and ingestion audit panels.	docs/screenshots/imd_evidence_15.png
IMG-16	Reviewer Dashboard	Reviewer audit dashboard empty-state prompt for historical analysis activation.	docs/screenshots/imd_evidence_16.png
IMG-17	Forecast Verification	Verification lab loading state while contingency matrices are calculated.	docs/screenshots/imd_evidence_17.png
IMG-18	Forecast Verification	Verification lab controls and calculation state in the Research & Soundings shell.	docs/screenshots/imd_evidence_18.png
IMG-19	Dataset Explorer	Dataset Explorer metric cards and seeded historical records log.	docs/screenshots/imd_evidence_19.png
IMG-20	Reviewer Dashboard	Reviewer audit dashboard prompt and reviewer workflow container.	docs/screenshots/imd_evidence_20.png
IMG-21	Forecast Verification	Verification lab with computed CSI, POD, FAR, BIAS, HSS and case inspection log.	docs/screenshots/imd_evidence_21.png
IMG-22	Forecast Verification	Computed verification lab with contingency matrix and operational decision advice.	docs/screenshots/imd_evidence_22.png
IMG-23	File Analysis Center	Uploaded dataset quality summary showing records, missing values, duplicates, quality score, and station coverage.	docs/screenshots/imd_evidence_23.png
IMG-24	Live Doppler Radar	IMD Visakhapatnam Cyclone Warning Centre radar console with operational metadata HUD.	docs/screenshots/imd_evidence_24.png
IMG-25	AI Prediction Engine	Regional threat queue, CAPE traceability, probability evolution, and recommended actions.	docs/screenshots/imd_evidence_25.png
IMG-26	AI Prediction Engine	Advanced diagnostics and scientific calibration blocks behind the operational summary.	docs/screenshots/imd_evidence_26.png
IMG-27	AI Prediction Engine	Composite severity, lifecycle timeline, and decision-weight calibration controls.	docs/screenshots/imd_evidence_27.png
IMG-28	Convective Analytics	Convective verification and threshold discovery hub with operational validation curves.	docs/screenshots/imd_evidence_28.png
IMG-29	Convective Analytics	Historical convective sounding log table from CWC archives.	docs/screenshots/imd_evidence_29.png
IMG-30	Operational Event Archive	Operational event archive cards for replaying historical severe weather scenarios.	docs/screenshots/imd_evidence_30.png
IMG-31	Skew-T Sounding Plot	Thermodynamic Skew-T Log-P sounding profile with metadata and station selection.	docs/screenshots/imd_evidence_31.png
IMG-32	Wyoming Sounding Data	University of Wyoming sounding archive view with source, cycle, cache age, and parameter decoder.	docs/screenshots/imd_evidence_32.png

ID	UI Area	Evidence Summary	Local File
IMG-33	Historical Thunderstorm Archive	Archive summary with latest observed thunderstorm event, record counts, and latest record metadata.	docs/screenshots/imd_evidence_33.png
IMG-34	Forecast Simulator	Custom sounding simulator with CAPE static-data warning and threshold comparison.	docs/screenshots/imd_evidence_34.png
IMG-35	Dataset Explorer	CAPE dynamicity audit table and latest archive metrics.	docs/screenshots/imd_evidence_35.png
IMG-36	Reviewer Dashboard	Reviewer audit dashboard with docket exports and reviewer verdict form.	docs/screenshots/imd_evidence_36.png
IMG-37	Research Navigation	Research & Soundings navigation with Start Here, About, archive, simulator, verification, and learning pages.	docs/screenshots/imd_evidence_37.png
IMG-38	Atmospheric Intelligence Center	Operational atmospheric intelligence center with station risk cards and interpretation.	docs/screenshots/imd_evidence_38.png
IMG-39	Atmospheric Intelligence Center	Compact atmospheric intelligence view answering current situation and recommended context.	docs/screenshots/imd_evidence_39.png
IMG-40	Severe Weather Watchdesk	Severe weather watchdesk with thunderstorm, heavy rain, lightning, squall, and stabilization indicators.	docs/screenshots/imd_evidence_40.png
IMG-41	Thermodynamic Lab	Thermodynamic lab with CAPE, CIN, LCL, LFC, EL, parcel trace narrative, interpretation, and actions.	docs/screenshots/imd_evidence_41.png
IMG-42	Climatology & Research Center	Climatology and research center with historical trust, Brier score, interpretation, and actions.	docs/screenshots/imd_evidence_42.png
IMG-43	Thunderstorm Forecast Simulator	Operational forecast workspace with file analysis center steps and forecast metric cards.	docs/screenshots/imd_evidence_43.png
IMG-44	Live Operational Nowcast Center	Nowcast center with coastal map markers and probability bars for thunderstorm, lightning, rain, and squall.	docs/screenshots/imd_evidence_44.png
IMG-45	Historical Thunderstorm Archive	Historical archive summary with file analysis center and latest thunderstorm evidence.	docs/screenshots/imd_evidence_45.png
IMG-46	Forecast Simulator	Simulator validation stages and CAPE static data warning evidence.	docs/screenshots/imd_evidence_46.png

Embedded Evidence Figures

IMG-01 - Historical Workbench

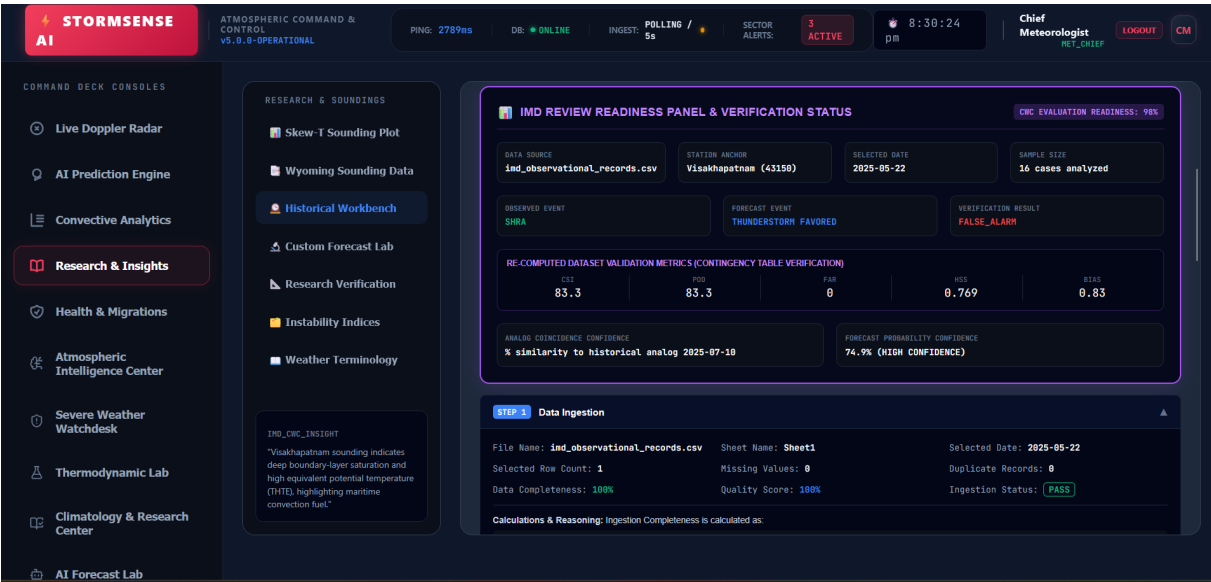
Historical thunderstorm analysis center with file/date/station selectors and IMD review readiness panel.



IMG-01 Historical Workbench

IMG-02 - Historical Workbench

Review readiness and verification status panel with contingency metrics and analog confidence.



IMG-02 Historical Workbench

IMG-03 - Historical Workbench

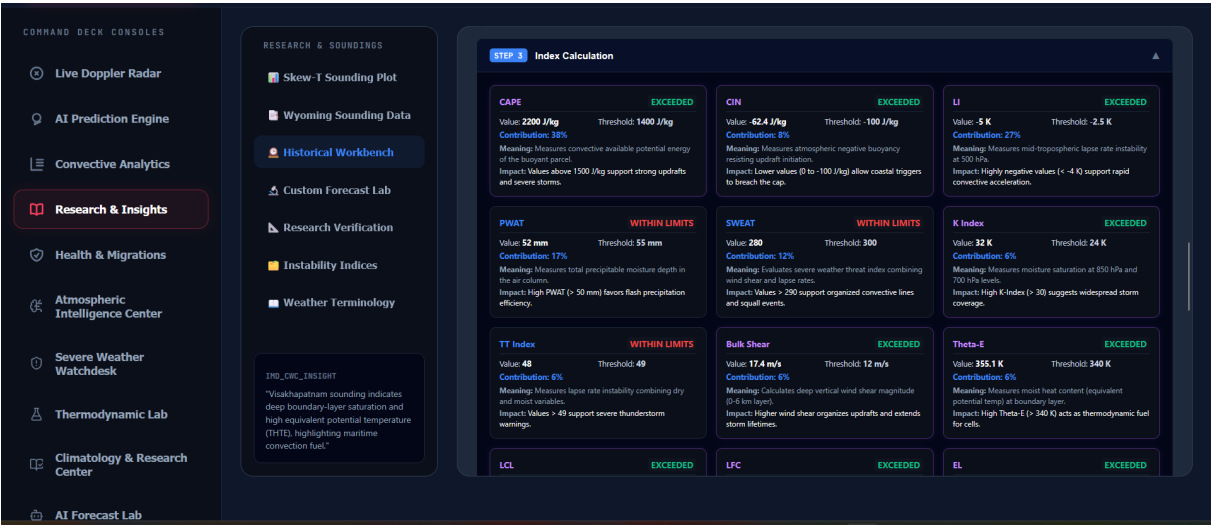
Pipeline evidence trail showing data ingestion and sounding parsing sections.



IMG-03 Historical Workbench

IMG-04 - Historical Workbench

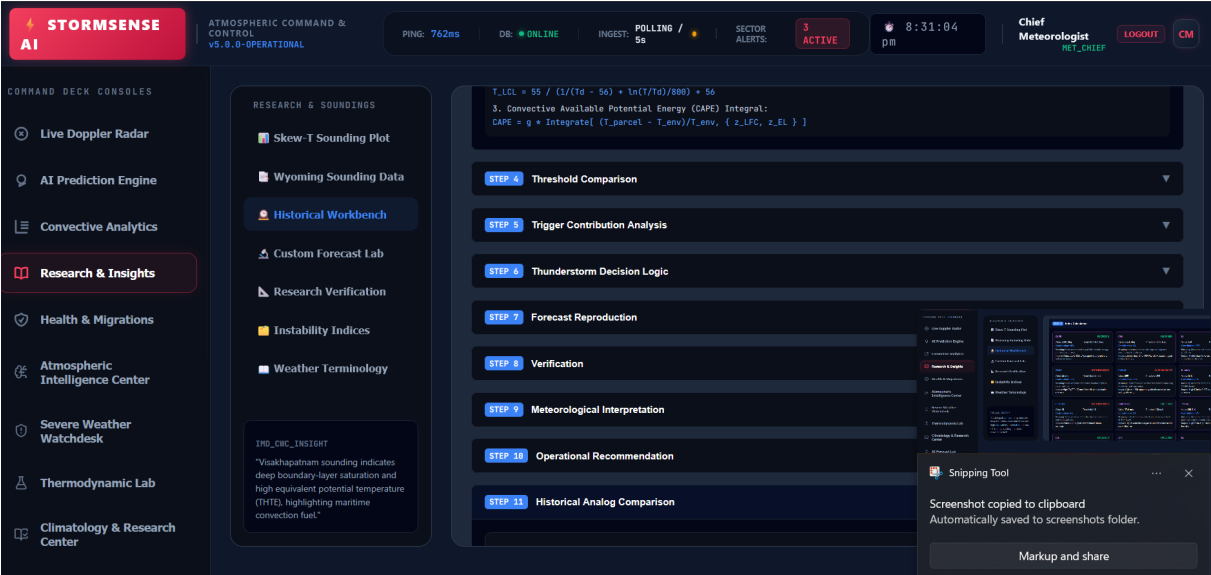
Index calculation cards for CAPE, CIN, LI, PWAT, SWEAT, K-index, shear, and theta-e.



IMG-04 Historical Workbench

IMG-05 - Historical Workbench

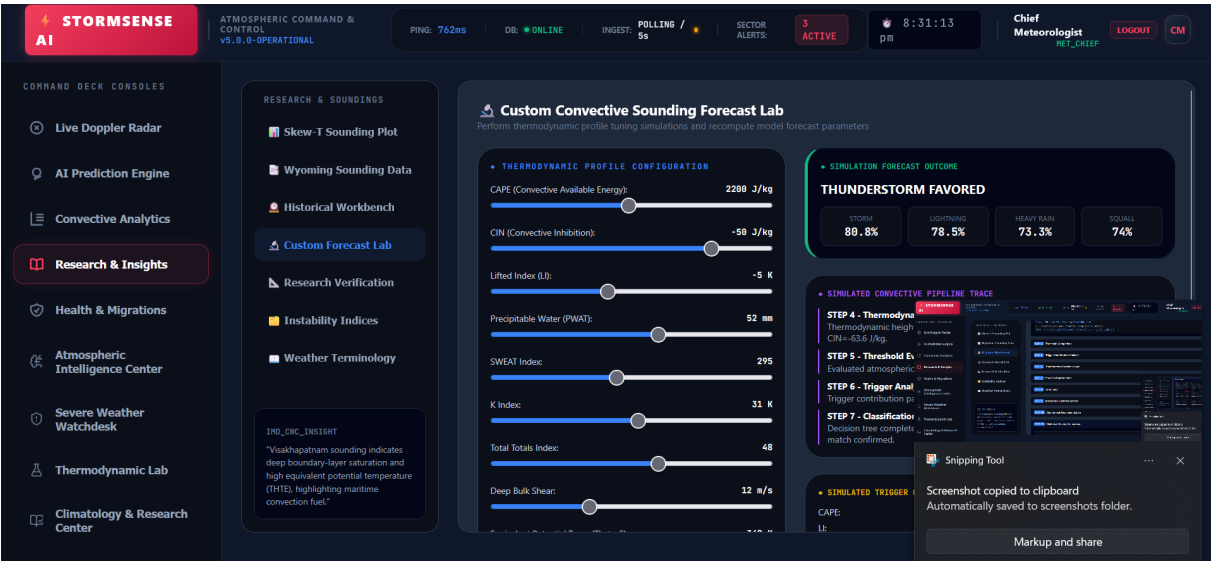
Collapsed workflow stages for threshold comparison, verification, interpretation, and recommendation.



IMG-05 Historical Workbench

IMG-06 - Forecast Simulator

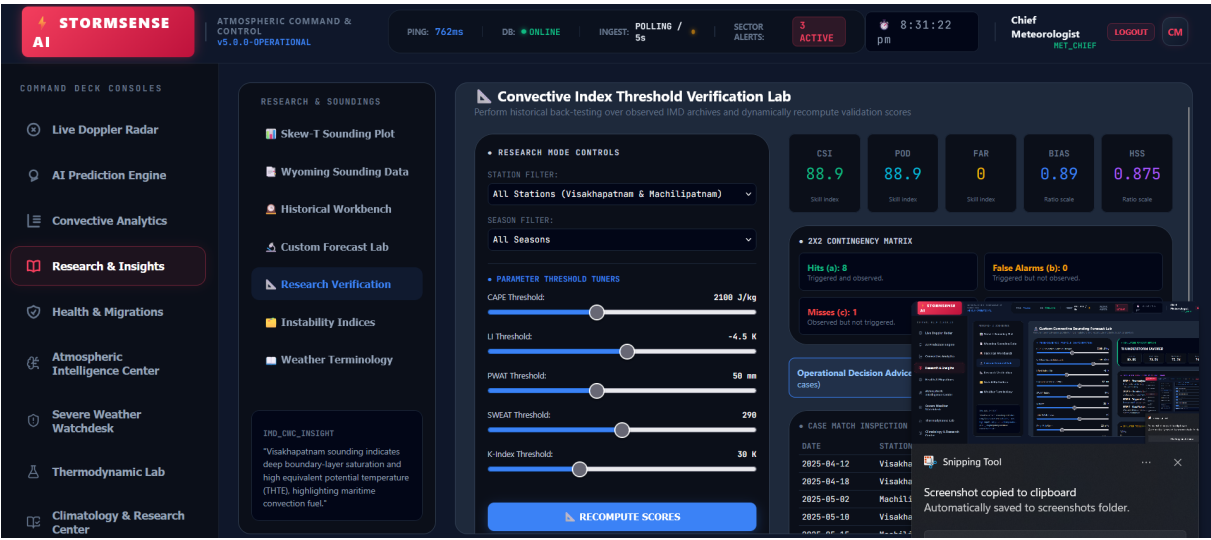
Custom convective sounding forecast lab with thermodynamic slider inputs and simulated forecast outcome.



IMG-06 Forecast Simulator

IMG-07 - Forecast Verification

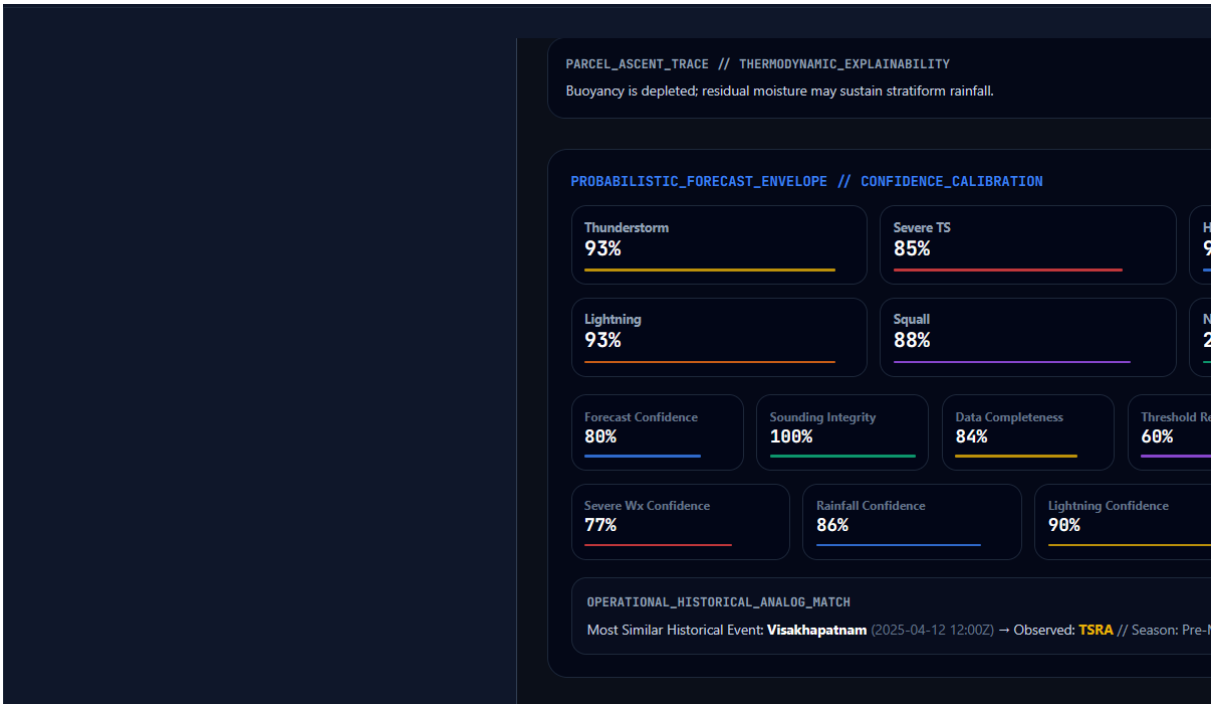
Convective index threshold verification lab with threshold controls and contingency metrics.



IMG-07 Forecast Verification

IMG-08 - AI Prediction Engine

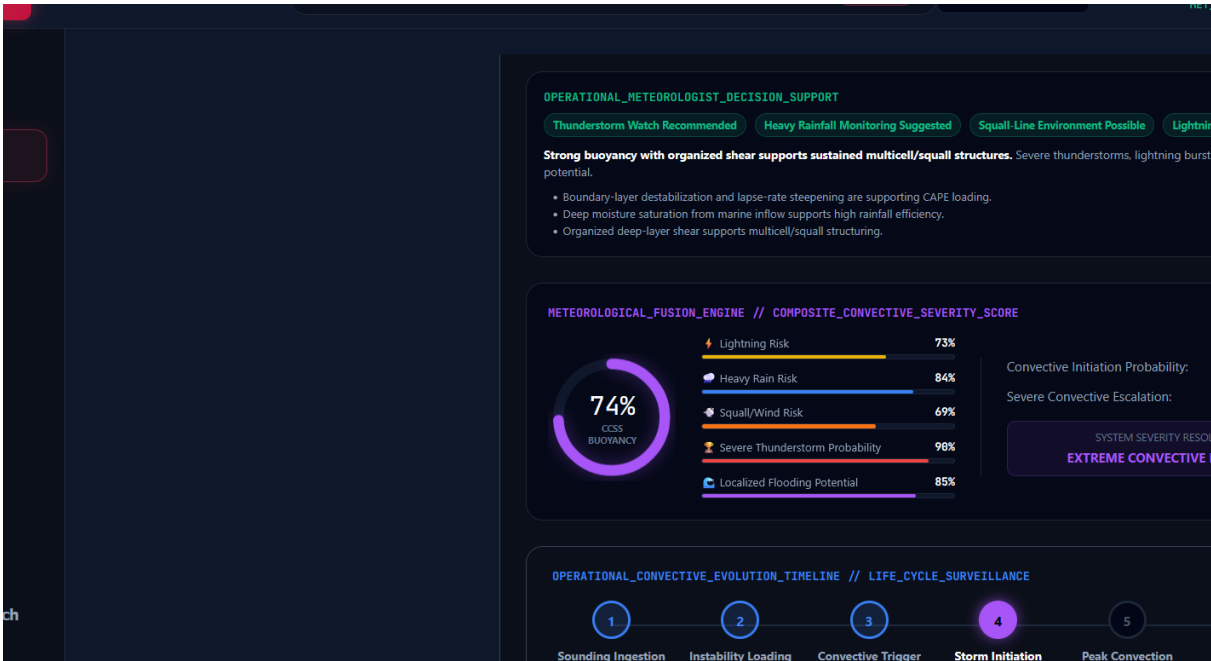
Probabilistic forecast envelope and operational analog match evidence.



IMG-08 AI Prediction Engine

IMG-09 - AI Prediction Engine

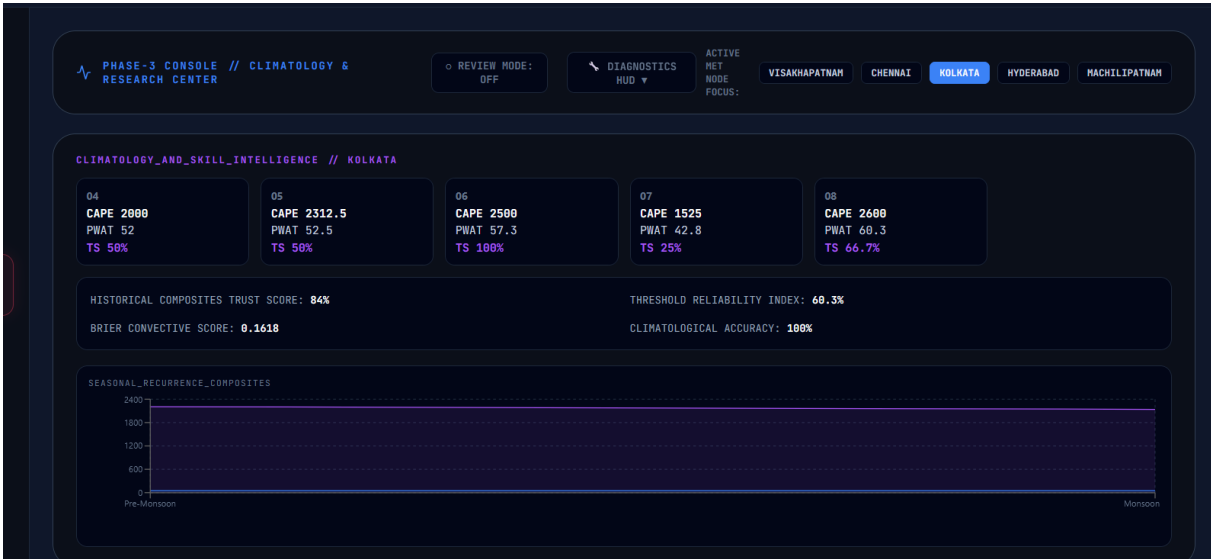
Operational meteorologist decision support and composite convective severity score.



IMG-09 AI Prediction Engine

IMG-10 - Climatology & Research Center

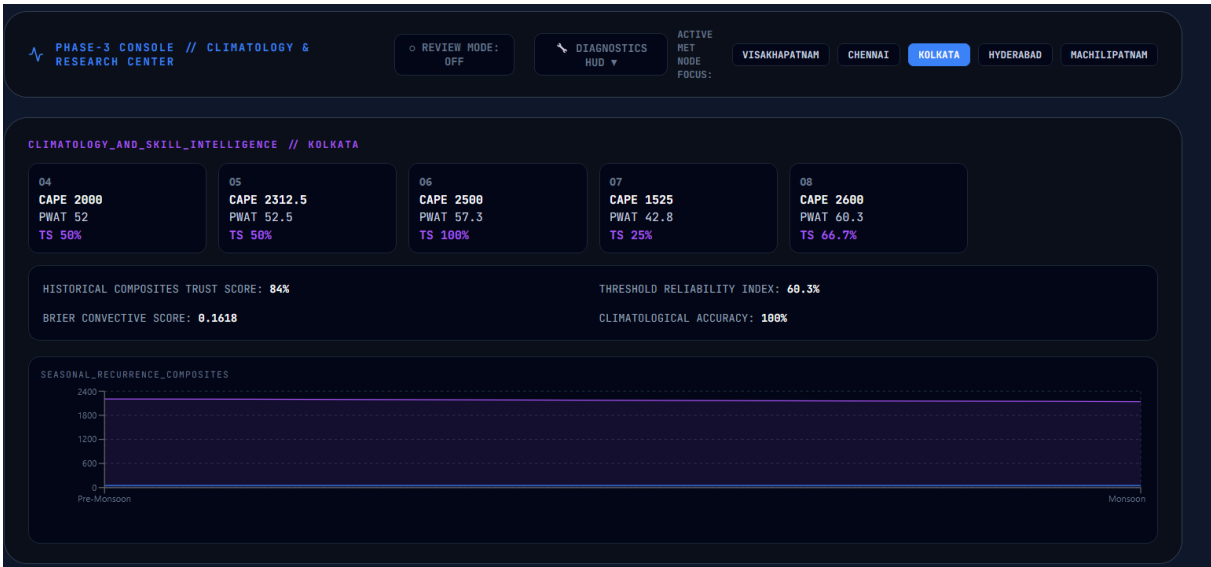
Climatology and skill intelligence station view with seasonal recurrence composites.



IMG-10 Climatology & Research Center

IMG-11 - Climatology & Research Center

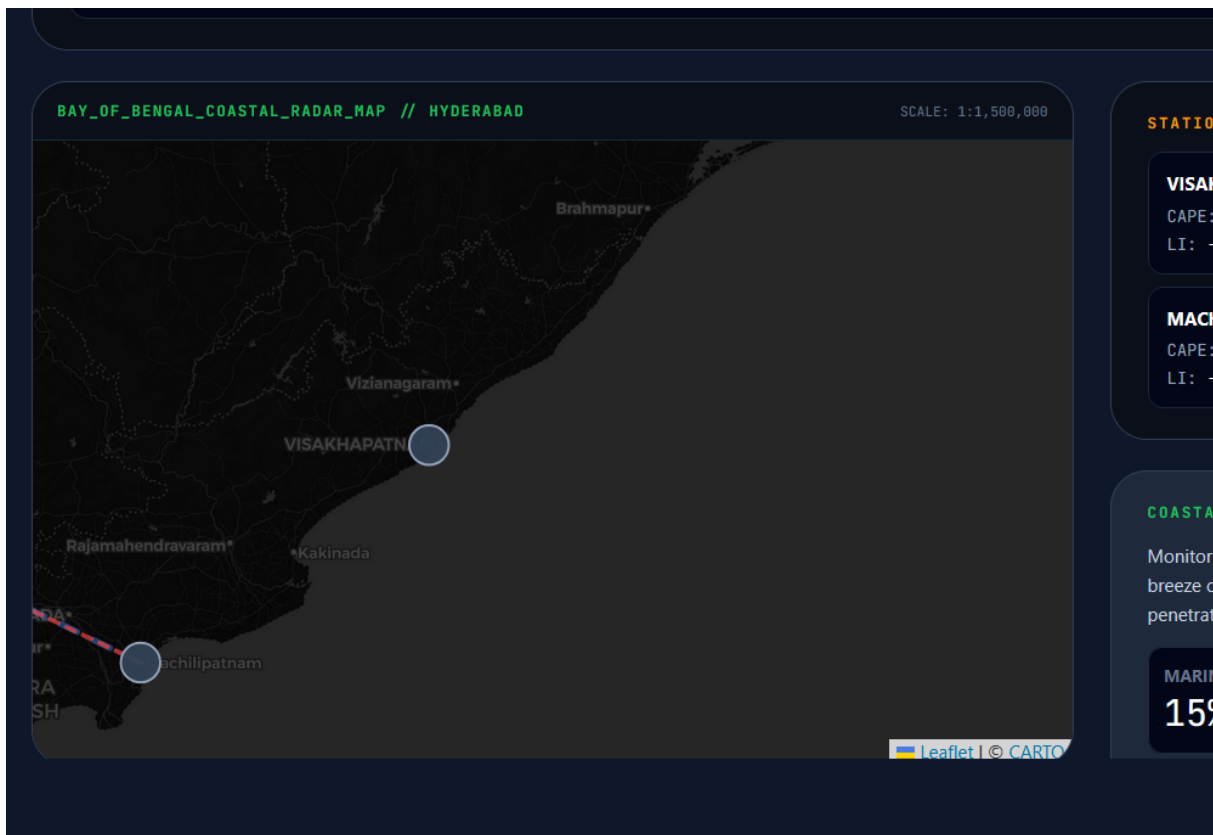
Expanded climatology view showing station composite metrics and chart area.



IMG-11 Climatology & Research Center

IMG-12 - Coastal Monitoring

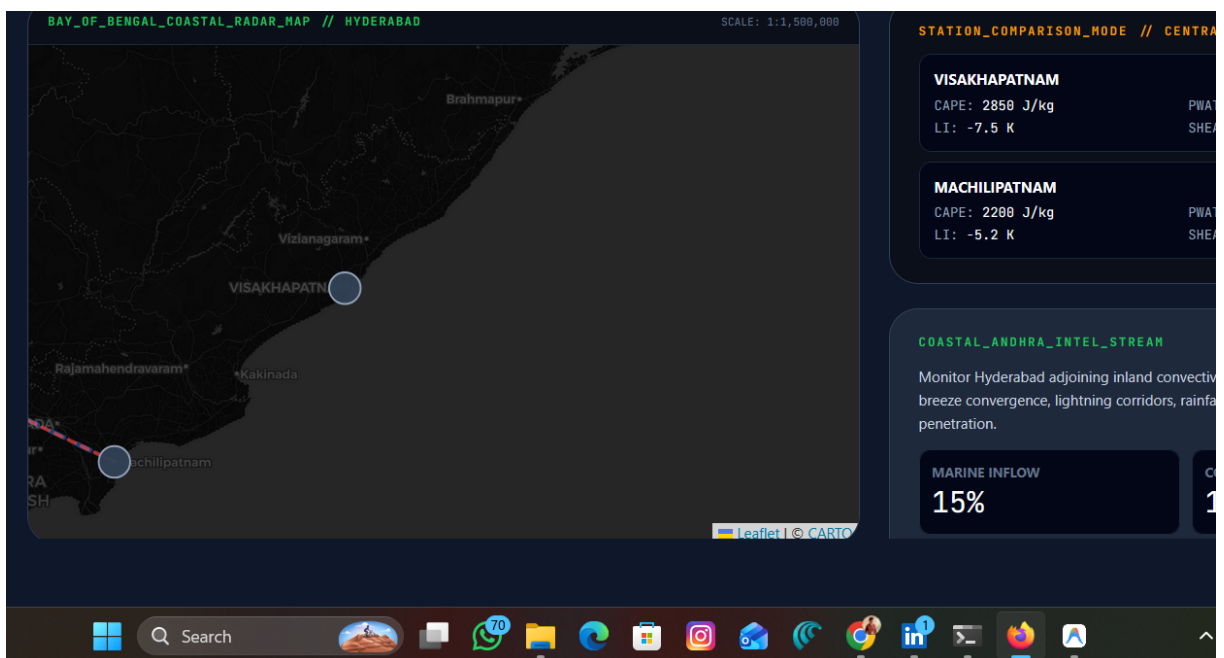
Bay of Bengal coastal radar map focused on north coastal Andhra and station comparison.



IMG-12 Coastal Monitoring

IMG-13 - Coastal Monitoring

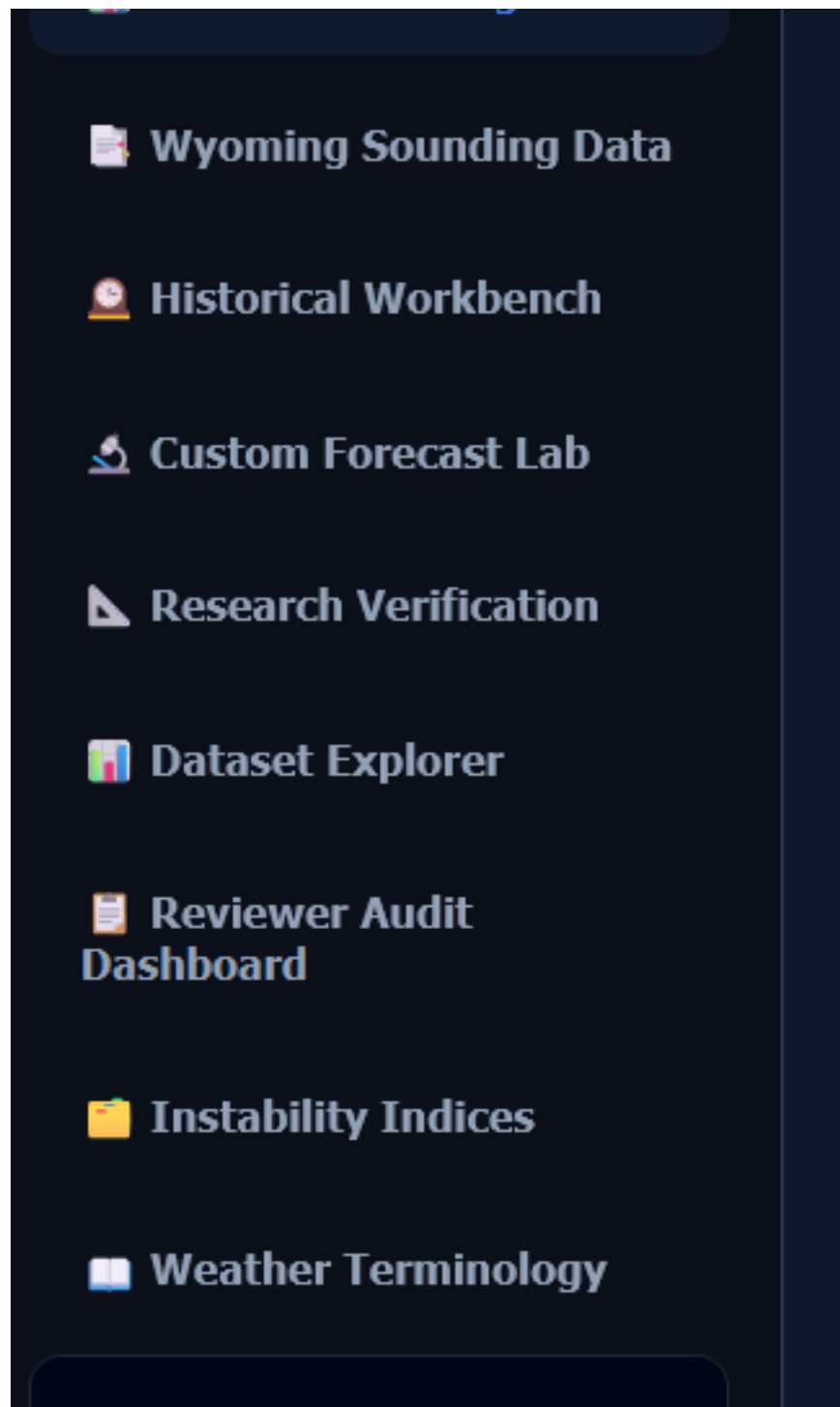
Coastal Andhra intelligence stream with marine inflow and corridor metrics.



IMG-13 Coastal Monitoring

IMG-14 - Research Navigation

Research & Soundings navigation stack with Dataset Explorer and Reviewer Audit Dashboard.



IMG-14 Research Navigation

Additional screenshots are catalogued above and retained in `docs/screenshots/`; the main dossier embeds the first 14 representative figures to keep the PDF/DOCX readable.

5. Research & Insights Documentation

Tab Code	Visible Page	Operational Function
START_HERE	START HERE	Reviewer onboarding center for data requirements, outputs, and common review flow.
ABOUT	ABOUT STORMSENSE AI	Operational reference page describing purpose, architecture, data sources, and workflows.
SKEW_T	Skew-T / Thermodynamic Lab	Sounding visualization and thermodynamic parameter interpretation.
WYOMING_DATA	Wyoming Data	Radiosonde ingest and cycle-aware sounding review.
HISTORICAL_WORKBENCH	Historical Thunderstorm Archive	Flagship investigation console for 2023, 2024, and 2025 archive records.
FORECAST_LAB	Thunderstorm Forecast Simulator	Historical and custom sounding forecast reproduction and dataset analysis.
RESEARCH_VERIFY	Forecast Verification	Contingency metrics, threshold testing, and reviewer verification workflows.
DATASET_EXPLORER	Historical Weather Database	Historical weather database and archive exploration workspace.
REVIEWER_DASHBOARD	IMD Review Dashboard	Reviewer docket, verdict, and operational review workspace.
INDEX_GLOSSARY	Thunderstorm Indices Guide	Guide to CAPE, LI, SWEAT, PWAT, K-index, CIN, LCL, LFC, EL, and verification metrics.
TERMINOLOGY	Meteorology Learning Center	Operational terminology and learning support for reviewers.

The Research & Insights module is the primary review and investigation area. It connects historical archive records, forecast simulation, verification, dataset upload analysis, thunderstorm registry, operational metadata, CAPE diagnostics, probability evolution, and reviewer workflow into a single operational research desk.

6. Meteorological Glossary

	Operational Interpretation
	Convective Available Potential Energy. High CAPE indicates buoyant energy available for deep convection. Operationally, rising CAPE with sufficient moisture and weak inhibition increases thunderstorm potential.
	Convective Inhibition. CIN represents the cap suppressing parcel ascent. Strong CIN can delay initiation even with high CAPE; weakening CIN can permit rapid thunderstorm development.
	Lifted Index. Negative LI indicates instability. Strong negative LI values support deep moist convection and severe updraft potential.
	Precipitable Water. High PWAT supports heavy rainfall efficiency, stratiform persistence, and flood susceptibility when forcing is present.
Thunderstorm	Severe Weather Threat Index. Combines instability, moisture, wind, and shear ingredients to indicate organized severe potential.
Wind Shear	Vector wind difference through a layer, commonly 0-6 km. Higher shear supports storm organization, propagation, and persistent cells.
	Lifted Condensation Level. Lower LCL indicates parcels reach saturation sooner, often supporting lower cloud bases and efficient convection.
	Level of Free Convection. The level where an air parcel becomes positively buoyant and can accelerate upward.
	Equilibrium Level. The upper level where the parcel loses buoyancy; higher EL often implies taller storms and stronger echo tops.
Equivalent	Equivalent potential temperature. High theta-e in low levels marks warm, moist air and instability corridors.
Thunderstorm Initiation	The transition from conditional instability to active deep convection after lifting, convergence, heating, or boundary forcing overcomes inhibition.
Conditional Instability	A thermodynamic profile capable of supporting deep convection when adequate lift and moisture are present.
	Critical Success Index. Measures event forecast accuracy while penalizing misses and false alarms.
	Probability of Detection. Fraction of observed events that were correctly forecast.
	False Alarm Ratio. Fraction of forecast events that did not occur; lower is better.
	Heidke Skill Score. Skill score relative to random chance, useful for categorical forecast verification.
	Forecast frequency bias. Values greater than 1 indicate overforecasting; less than 1 indicates underforecasting.

7. Database Documentation

StormSense AI uses Supabase PostgreSQL for production persistence. The schema is defined in `supabase\_schema.sql` and initialized/accessed through SQLAlchemy helpers.

Table	Columns
roles	id, name, description, created_at
users	id, name, email, password, role, created_at
profiles	id, user_id, auth_user_id, display_name, station_scope, created_at, updated_at
audit_login	id, user_id, email, status, detail, created_at
thunderstorm_forecasts	id, station, station_code, cape, lifted_index, sweat_index, k_index, pwat ...
historical_observations	id, record_date, observation_time, station, station_code, observed_event, source_file, source_sheet ...
historical_records	id, record_date, observation_time, station, station_code, observed_event, source_file, source_sheet ...
thunderstorm_registry	id, record_date, observation_time, station, station_code, observed_event, thunderstorm, nwx ...
verification_results	id, record_date, station, observed_event, forecast_result, verification_result, csi, hss ...
uploaded_datasets	id, file_name, storage_bucket, storage_path, total_records, thunderstorm_records, nwx_records, severe_storm_records ...
reviewer_dashboard_records	id, reviewer_id, docket_id, station, record_date, verdict, recommended_action, comments ...
audit_logs	id, action, status, detail, created_at

```
erDiagram
    users ||--o{ profiles : owns
    users ||--o{ audit_login : records
    thunderstorm_forecasts ||--o{ verification_results : validates
    historical_observations ||--o{ thunderstorm_registry : classifies
    uploaded_datasets ||--o{ thunderstorm_registry : generates
    reviewer_dashboard_records }o--|| users : reviewed_by
```

Table-by-Table Supabase Notes

roles

Role catalog for operational user classification.

Column Definition
id BIGINT GENERATED ALWAYS AS IDENTITY PRIMARY KEY
name VARCHAR(100) NOT NULL UNIQUE
description TEXT
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP

Review validation: confirm row count after migration, primary key behavior, nullable fields, and related frontend workflow visibility.

users

Persisted application users migrated from the MySQL Workbench export and used by FastAPI authentication.

Column Definition
id BIGINT GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY
name VARCHAR(255) NOT NULL

Column Definition
email VARCHAR(255) NOT NULL UNIQUE
password VARCHAR(255) NOT NULL
role VARCHAR(100) NOT NULL DEFAULT 'MET_CHIEF'
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP

Review validation: confirm row count after migration, primary key behavior, nullable fields, and related frontend workflow visibility.

profiles

Optional profile extension table linked to users and future Supabase auth identifiers.

Column Definition
id BIGINT GENERATED ALWAYS AS IDENTITY PRIMARY KEY
user_id BIGINT REFERENCES users(id) ON DELETE CASCADE
auth_user_id UUID
display_name VARCHAR(255)
station_scope TEXT
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP

Review validation: confirm row count after migration, primary key behavior, nullable fields, and related frontend workflow visibility.

audit_login

Login audit trail recording successful and failed authentication attempts.

Column Definition
id BIGINT GENERATED ALWAYS AS IDENTITY PRIMARY KEY
user_id BIGINT REFERENCES users(id) ON DELETE SET NULL
email VARCHAR(255)
status VARCHAR(40) NOT NULL CHECK (status IN ('SUCCESS', 'FAILED', 'LOCKED', 'REVOKED'))
detail TEXT
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP

Review validation: confirm row count after migration, primary key behavior, nullable fields, and related frontend workflow visibility.

thunderstorm_forecasts

Primary forecast persistence table containing station, indices, forecast class, probability, and created timestamp.

Column Definition
id BIGINT GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY
station VARCHAR(255)

Column Definition
station_code VARCHAR(255)
cape DOUBLE PRECISION
lifted_index DOUBLE PRECISION
sweat_index DOUBLE PRECISION
k_index DOUBLE PRECISION
pwat DOUBLE PRECISION
forecast VARCHAR(255)
storm_probability INTEGER CHECK (storm_probability IS NULL OR storm_probability BETWEEN 0 AND 100)
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP

Review validation: confirm row count after migration, primary key behavior, nullable fields, and related frontend workflow visibility.

historical_observations

Normalized historical observation records used by archive and verification workflows.

Column Definition
id BIGINT GENERATED ALWAYS AS IDENTITY PRIMARY KEY
record_date DATE
observation_time VARCHAR(40)
station VARCHAR(255)
station_code VARCHAR(50)
observed_event VARCHAR(100)
source_file TEXT
source_sheet TEXT
raw_json JSONB
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP

Review validation: confirm row count after migration, primary key behavior, nullable fields, and related frontend workflow visibility.

historical_records

Historical record mirror/compatibility table for archive workflows.

Column Definition
id BIGINT GENERATED ALWAYS AS IDENTITY PRIMARY KEY
record_date DATE
observation_time VARCHAR(40)
station VARCHAR(255)
station_code VARCHAR(50)
observed_event VARCHAR(100)

Column Definition
source_file TEXT
source_sheet TEXT
raw_json JSONB
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP

Review validation: confirm row count after migration, primary key behavior, nullable fields, and related frontend workflow visibility.

thunderstorm_registry

File/archive derived event registry with thunderstorm, NWX, severe storm, forecast, verification, and season flags.

Column Definition
id BIGINT GENERATED ALWAYS AS IDENTITY PRIMARY KEY
record_date DATE
observation_time VARCHAR(40)
station VARCHAR(255)
station_code VARCHAR(50)
observed_event VARCHAR(100)
thunderstorm BOOLEAN DEFAULT FALSE
nwx BOOLEAN DEFAULT FALSE
severe_storm BOOLEAN DEFAULT FALSE
forecast_result VARCHAR(255)
verification_result VARCHAR(100)
season VARCHAR(40)
source_dataset TEXT
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP

Review validation: confirm row count after migration, primary key behavior, nullable fields, and related frontend workflow visibility.

verification_results

Forecast verification outputs, including CSI, HSS, POD, FAR, and BIAS.

Column Definition
id BIGINT GENERATED ALWAYS AS IDENTITY PRIMARY KEY
record_date DATE
station VARCHAR(255)
observed_event VARCHAR(100)
forecast_result VARCHAR(255)
verification_result VARCHAR(100)
csi DOUBLE PRECISION

Column Definition
hss DOUBLE PRECISION
pod DOUBLE PRECISION
far DOUBLE PRECISION
bias DOUBLE PRECISION
source_dataset TEXT
raw_json JSONB
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP

Review validation: confirm row count after migration, primary key behavior, nullable fields, and related frontend workflow visibility.

uploaded_datasets

File Analysis Center persistence for upload metadata, quality score, and analysis JSON.

Column Definition
id BIGINT GENERATED ALWAYS AS IDENTITY PRIMARY KEY
file_name TEXT NOT NULL
storage_bucket TEXT
storage_path TEXT
total_records INTEGER DEFAULT 0
thunderstorm_records INTEGER DEFAULT 0
nwx_records INTEGER DEFAULT 0
severe_storm_records INTEGER DEFAULT 0
quality_score INTEGER DEFAULT 0 CHECK (quality_score BETWEEN 0 AND 100)
analysis_json JSONB
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP

Review validation: confirm row count after migration, primary key behavior, nullable fields, and related frontend workflow visibility.

reviewer_dashboard_records

Reviewer docket, verdict, comments, station, date, and export metadata.

Column Definition
id BIGINT GENERATED ALWAYS AS IDENTITY PRIMARY KEY
reviewer_id TEXT
docket_id TEXT
station VARCHAR(255)
record_date DATE
verdict VARCHAR(100)
recommended_action TEXT

Column Definition
comments TEXT
export_storage_path TEXT
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP

Review validation: confirm row count after migration, primary key behavior, nullable fields, and related frontend workflow visibility.

audit_logs

General operational audit log for migration and backend activity.

Column Definition
id BIGINT GENERATED ALWAYS AS IDENTITY PRIMARY KEY
action VARCHAR(100) NOT NULL
status VARCHAR(40) NOT NULL
detail TEXT
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP

Review validation: confirm row count after migration, primary key behavior, nullable fields, and related frontend workflow visibility.

8. API Documentation

Full API reference is also generated separately in `API\_Reference.md`. Summary of discovered routes (54):

Method	Route	Handler	Source
POST	/auth/signup	signup	backend/main.py:575
POST	/auth/login	login	backend/main.py:613
GET	/auth/me	auth_me	backend/main.py:703
POST	/cwc/override	trigger_override	backend/main.py:711
POST	/cwc/clear-override	clear_override	backend/main.py:732
POST	/cwc/cycle	update_cycle	backend/main.py:754
GET	/cwc/cycle	get_cycle	backend/main.py:765
GET	/	home	backend/main.py:1558
GET	/forecast	forecast	backend/main.py:1574
GET	/history	history	backend/main.py:1592
GET	/trend-analysis	trend_analysis	backend/main.py:1636
GET	/storm-escalation	storm_escalation	backend/main.py:1670
WEBSOCKET	/stream/atmospheric	stream_atmospheric	backend/main.py:1717
GET	/system-status	system_status	backend/main.py:1769
GET	/cwc/correlation	get_cwc_correlation	backend/main.py:1792
GET	/cwc/optimization	get_cwc_optimization	backend/main.py:1796
GET	/cwc/verification	get_cwc_verification	backend/main.py:1800
GET	/cwc/thresholds	get_cwc_thresholds	backend/main.py:1804
GET	/cwc/index-catalog	get_cwc_index_catalog	backend/main.py:1817
GET	/cwc/seasonal-analysis	get_cwc_seasonal_analysis	backend/main.py:1821
GET	/cwc/replay-cases	get_cwc_replay_cases	backend/main.py:1825
GET	/cwc/observations	get_cwc_observations	backend/main.py:1829
GET	/cwc/observational-analytics	get_cwc_observational_analytics	backend/main.py:1833
GET	/cwc/probabilistic-forecast	get_cwc_probabilistic_forecast	backend/main.py:1837
GET	/cwc/thermo-diagnostics	get_cwc_thermo_diagnostics	backend/main.py:1845
GET	/cwc/verification-advanced	get_cwc_verification_advanced	backend/main.py:1901
GET	/cwc/ml-ready-dataset	get_cwc_ml_ready_dataset	backend/main.py:1905
GET	/cwc/climatology	get_cwc_climatology	backend/main.py:1909
GET	/cwc/decision-support	get_cwc_decision_support	backend/main.py:1913
GET	/cwc/operational-bulletins	get_cwc_operational_bulletins	backend/main.py:1967
GET	/cwc/radar-sounding-fusion	get_cwc_radar_sounding_fusion	backend/main.py:1974
GET	/cwc/ai-forecast-intelligence	get_cwc_ai_forecast_intelligence	backend/main.py:1993
GET	/cwc/coastal-andhra-intelligence	get_cwc_coastal_andhra_intelligence	backend/main.py:2018

Method	Route	Handler	Source
GET	/cwc/verification-rolling	get_cwc_verification_rolling	backend/main.py:2042
GET	/cwc/historical-observations	get_cwc_historical_observations	backend/main.py:2046
GET	/cwc/export/csv	export_cwc_csv	backend/main.py:2050
GET	/cwc/export/json	export_cwc_json	backend/main.py:2059
GET	/cwc/ml-pipeline	get_cwc_ml_pipeline	backend/main.py:2063
GET	/cwc/sounding-raw/{station_code}	get_cwc_sounding_raw	backend/main.py:2068
GET	/cwc/threshold-research	get_cwc_threshold_research	backend/main.py:2090
GET	/cwc/cape-traceability	get_cwc_cape_traceability	backend/main.py:2131
GET	/cwc/analog	get_cwc_analog	backend/main.py:2176
GET	/cwc/forecast-evolution	get_cwc_forecast_evolution	backend/main.py:2193
GET	/cwc/district-impact	get_cwc_district_impact	backend/main.py:2203
GET	/cwc/historical-files	get_cwc_historical_files	backend/main.py:2243
GET	/cwc/historical-dates	get_cwc_historical_dates	backend/main.py:2440
GET	/cwc/historical-analysis	get_cwc_historical_analysis	backend/main.py:2489
POST	/cwc/upload-sounding	upload_sounding	backend/main.py:2802
POST	/cwc/analyze-historical-dataset	analyze_historical_dataset	backend/main.py:2859
GET	/cwc/latest-file-analyzed	get_latest_file_analyzed	backend/main.py:3082
POST	/cwc/custom-sounding-analysis	custom_sounding_analysis	backend/main.py:3086
POST	/cwc/research-verification	post_cwc_research_verification	backend/main.py:3167
GET	/cwc/export/analysis	export_cwc_analysis	backend/main.py:3184
GET	{full_path:path}	serve_frontend_spa	backend/main.py:3265

Complete API Endpoint Detail

POST /auth/signup

Source reference: `backend/main.py:575` handler `signup`.

Purpose: Register an operational user in the StormSense authentication store.

Input: JSON request body matching the route's Pydantic model or route-specific payload fields.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

POST /auth/login

Source reference: `backend/main.py:613` handler `login`.

Purpose: Authenticate a reviewer/operator and return a bearer token plus user profile.

Input: JSON request body matching the route's Pydantic model or route-specific payload fields.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /auth/me

Source reference: `backend/main.py:703` handler `auth\_me`.

Purpose: Return the authenticated operator profile for session validation.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: Bearer token required

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

POST /cwc/override

Source reference: `backend/main.py:711` handler `trigger\_override`.

Purpose: Expose trigger override workflow output for the operational workstation.

Input: JSON request body matching the route's Pydantic model or route-specific payload fields.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: Bearer token required

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

POST /cwc/clear-override

Source reference: `backend/main.py:732` handler `clear\_override`.

Purpose: Expose clear override workflow output for the operational workstation.

Input: JSON request body matching the route's Pydantic model or route-specific payload fields.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: Bearer token required

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

POST /cwc/cycle

Source reference: `backend/main.py:754` handler `update\_cycle`.

Purpose: Expose update cycle workflow output for the operational workstation.

Input: JSON request body matching the route's Pydantic model or route-specific payload fields.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: Bearer token required

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /cwc/cycle

Source reference: `backend/main.py:765` handler `get\_cycle`.

Purpose: Expose get cycle workflow output for the operational workstation.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /

Source reference: `backend/main.py:1558` handler `home`.

Purpose: Expose home workflow output for the operational workstation.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /forecast

Source reference: `backend/main.py:1574` handler `forecast`.

Purpose: Generate live station forecasts using sounding, thermodynamic, probability, and decision-support engines.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /history

Source reference: `backend/main.py:1592` handler `history`.

Purpose: Return recent persisted thunderstorm forecast records.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /trend-analysis

Source reference: `backend/main.py:1636` handler `trend\_analysis`.

Purpose: Return station-level trend analysis for convective evolution monitoring.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /storm-escalation

Source reference: `backend/main.py:1670` handler `storm\_escalation`.

Purpose: Return severe escalation records for high-risk station monitoring.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

WEBSOCKET /stream/atmospheric

Source reference: `backend/main.py:1717` handler `stream\_atmospheric`.

Purpose: Stream atmospheric forecast, trend, escalation, and cycle metadata over websocket.

Input: Websocket connection request from the frontend telemetry client.

Output: Streaming JSON frames containing forecasts, trends, escalations, and cycle metadata.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /system-status

Source reference: `backend/main.py:1769` handler `system\_status`.

Purpose: Return backend, database, websocket, and runtime health status.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /cwc/correlation

Source reference: `backend/main.py:1792` handler `get\_cwc\_correlation`.

Purpose: Expose get cwc correlation workflow output for the operational workstation.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /cwc/optimization

Source reference: `backend/main.py:1796` handler `get\_cwc\_optimization`.

Purpose: Expose get cwc optimization workflow output for the operational workstation.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /cwc/verification

Source reference: `backend/main.py:1800` handler `get\_cwc\_verification`.

Purpose: Expose get cwc verification workflow output for the operational workstation.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /cwc/thresholds

Source reference: `backend/main.py:1804` handler `get\_cwc\_thresholds`.

Purpose: Expose get cwc thresholds workflow output for the operational workstation.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /cwc/index-catalog

Source reference: `backend/main.py:1817` handler `get\_cwc\_index\_catalog`.

Purpose: Expose get cwc index catalog workflow output for the operational workstation.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /cwc/seasonal-analysis

Source reference: `backend/main.py:1821` handler `get\_cwc\_seasonal\_analysis`.

Purpose: Expose get cwc seasonal analysis workflow output for the operational workstation.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /cwc/replay-cases

Source reference: `backend/main.py:1825` handler `get\_cwc\_replay\_cases`.

Purpose: Expose get cwc replay cases workflow output for the operational workstation.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /cwc/observations

Source reference: `backend/main.py:1829` handler `get\_cwc\_observations`.

Purpose: Expose get cwc observations workflow output for the operational workstation.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /cwc/observational-analytics

Source reference: `backend/main.py:1833` handler `get\_cwc\_observational\_analytics`.

Purpose: Expose get cwc observational analytics workflow output for the operational workstation.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /cwc/probabilistic-forecast

Source reference: `backend/main.py:1837` handler `get\_cwc\_probabilistic\_forecast`.

Purpose: Expose get cwc probabilistic forecast workflow output for the operational workstation.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /cwc/thermo-diagnostics

Source reference: `backend/main.py:1845` handler `get\_cwc\_thermo\_diagnostics`.

Purpose: Expose get cwc thermo diagnostics workflow output for the operational workstation.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /cwc/verification-advanced

Source reference: `backend/main.py:1901` handler `get\_cwc\_verification\_advanced`.

Purpose: Expose get cwc verification advanced workflow output for the operational workstation.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /cwc/ml-ready-dataset

Source reference: `backend/main.py:1905` handler `get\_cwc\_ml\_ready\_dataset`.

Purpose: Expose get cwc ml ready dataset workflow output for the operational workstation.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /cwc/climatology

Source reference: `backend/main.py:1909` handler `get\_cwc\_climatology`.

Purpose: Expose get cwc climatology workflow output for the operational workstation.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /cwc/decision-support

Source reference: `backend/main.py:1913` handler `get\_cwc\_decision\_support`.

Purpose: Expose get cwc decision support workflow output for the operational workstation.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /cwc/operational-bulletins

Source reference: `backend/main.py:1967` handler `get\_cwc\_operational\_bulletins`.

Purpose: Generate operational bulletin products from current forecast rows.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /cwc/radar-sounding-fusion

Source reference: `backend/main.py:1974` handler `get\_cwc\_radar\_sounding\_fusion`.

Purpose: Expose get cwc radar sounding fusion workflow output for the operational workstation.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /cwc/ai-forecast-intelligence

Source reference: `backend/main.py:1993` handler `get\_cwc\_ai\_forecast\_intelligence`.

Purpose: Expose get cwc ai forecast intelligence workflow output for the operational workstation.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /cwc/coastal-andhra-intelligence

Source reference: `backend/main.py:2018` handler `get\_cwc\_coastal\_andhra\_intelligence`.

Purpose: Expose get cwc coastal andhra intelligence workflow output for the operational workstation.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /cwc/verification-rolling

Source reference: `backend/main.py:2042` handler `get\_cwc\_verification\_rolling`.

Purpose: Expose get cwc verification rolling workflow output for the operational workstation.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /cwc/historical-observations

Source reference: `backend/main.py:2046` handler `get\_cwc\_historical\_observations`.

Purpose: Expose get cwc historical observations workflow output for the operational workstation.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /cwc/export/csv

Source reference: `backend/main.py:2050` handler `export\_cwc\_csv`.

Purpose: Expose export cwc csv workflow output for the operational workstation.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: CSV, JSON, or analysis export response depending on route and query parameters.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /cwc/export/json

Source reference: `backend/main.py:2059` handler `export\_cwc\_json`.

Purpose: Expose export cwc json workflow output for the operational workstation.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: CSV, JSON, or analysis export response depending on route and query parameters.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /cwc/ml-pipeline

Source reference: `backend/main.py:2063` handler `get\_cwc\_ml\_pipeline`.

Purpose: Expose get cwc ml pipeline workflow output for the operational workstation.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /cwc/sounding-raw/{station_code}

Source reference: `backend/main.py:2068` handler `get\_cwc\_sounding\_raw`.

Purpose: Expose get cwc sounding raw workflow output for the operational workstation.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: Raw sounding text plus sounding metadata for source traceability.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /cwc/threshold-research

Source reference: `backend/main.py:2090` handler `get\_cwc\_threshold\_research`.

Purpose: Expose get cwc threshold research workflow output for the operational workstation.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /cwc/cape-traceability

Source reference: `backend/main.py:2131` handler `get\_cwc\_cape\_traceability`.

Purpose: Return station CAPE traceability timeline and static-data warnings.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /cwc/analog

Source reference: `backend/main.py:2176` handler `get\_cwc\_analog`.

Purpose: Expose get cwc analog workflow output for the operational workstation.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /cwc/forecast-evolution

Source reference: `backend/main.py:2193` handler `get\_cwc\_forecast\_evolution`.

Purpose: Expose get cwc forecast evolution workflow output for the operational workstation.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /cwc/district-impact

Source reference: `backend/main.py:2203` handler `get\_cwc\_district\_impact`.

Purpose: Return district impact intelligence for coastal Andhra monitoring.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /cwc/historical-files

Source reference: `backend/main.py:2243` handler `get\_cwc\_historical\_files`.

Purpose: Expose get cwc historical files workflow output for the operational workstation.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /cwc/historical-dates

Source reference: `backend/main.py:2440` handler `get\_cwc\_historical\_dates`.

Purpose: Return the historical archive index used by search, registry, and reviewer workflows.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /cwc/historical-analysis

Source reference: `backend/main.py:2489` handler `get\_cwc\_historical\_analysis`.

Purpose: Return a full historical case analysis with forecast reproduction and verification fields.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

POST /cwc/upload-sounding

Source reference: `backend/main.py:2802` handler `upload\_sounding`.

Purpose: Expose upload sounding workflow output for the operational workstation.

Input: Multipart file upload. Supported operational dataset formats include CSV, XLS, XLSX, radiosonde text, and historical sounding datasets where implemented.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

POST /cwc/analyze-historical-dataset

Source reference: `backend/main.py:2859` handler `analyze\_historical\_dataset`.

Purpose: Analyze uploaded CSV/XLS/XLSX historical datasets and build file-based registry output.

Input: Multipart file upload. Supported operational dataset formats include CSV, XLS, XLSX, radiosonde text, and historical sounding datasets where implemented.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /cwc/latest-file-analyzed

Source reference: `backend/main.py:3082` handler `get\_latest\_file\_analyzed`.

Purpose: Return the latest uploaded file analysis metadata for dashboard reference.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

POST /cwc/custom-sounding-analysis

Source reference: `backend/main.py:3086` handler `custom\_sounding\_analysis`.

Purpose: Expose custom sounding analysis workflow output for the operational workstation.

Input: JSON request body matching the route's Pydantic model or route-specific payload fields.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

POST /cwc/research-verification

Source reference: `backend/main.py:3167` handler `post\_cwc\_research\_verification`.

Purpose: Expose post cwc research verification workflow output for the operational workstation.

Input: JSON request body matching the route's Pydantic model or route-specific payload fields.

Output: JSON response containing operational data, metadata, diagnostics, or verification results.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

GET /cwc/export/analysis

Source reference: `backend/main.py:3184` handler `export\_cwc\_analysis`.

Purpose: Expose export cwc analysis workflow output for the operational workstation.

Input: Optional query parameters where declared by the route definition; otherwise no request body.

Output: CSV, JSON, or analysis export response depending on route and query parameters.

Authentication: No bearer token required for standard dashboard/review retrieval unless deployment policy adds external access control.

Operational review note: validate this endpoint during IMD/CWC review if its paired frontend module is demonstrated. Confirm HTTP status, payload shape, source metadata, and user-visible module behavior.

9. Deployment Documentation

Full deployment guide is generated separately in `Deployment_Guide.md`. The current production direction is a single Render Web Service. `render.yaml` builds backend requirements, builds the Vite frontend, and starts `uvicorn backend.main:app`.

10. Testing Documentation

Test Area	Validation Method	Expected Result
Frontend build	`npm install`; `npm run build`	`frontend/dist` generated successfully.
Backend compile	`python -m py_compile backend/main.py`	No syntax errors.
Backend import	`python -c "import backend.main"`	Application imports without ModuleNotFoundError.
API smoke	GET `/system-status`, `/history`, `/cwc/historical-dates`, `/forecast`	HTTP 200 JSON responses.
SPA refresh	GET `/dashboard`, `/research`, `/archive`, `/forecast` with `Accept: text/html`	HTTP 200 React HTML shell.
Upload workflow	POST `/cwc/analyze-historical-dataset` with CSV/XLS/XLSX	Dataset quality summary and registry returned.
Verification workflow	Run threshold research and contingency analysis	CSI, HSS, POD, FAR, BIAS returned.
Review mode	Enable IMD Review Mode	Simplified review deck with only reviewer-critical fields.

Detailed Test Scenarios

Authentication Testing

POST `/auth/login` with a known reviewer account, verify token issuance, store token in frontend local storage, call `/auth/me`, then logout and confirm protected actions are unavailable.

Expected outcome: pass/fail result must be recorded with endpoint, UI location, dataset/source, and reviewer-visible evidence.

Historical Archive Testing

Open Research & Insights -> Historical Thunderstorm Archive. Confirm 2023, 2024, and 2025 records are visible in summaries, latest thunderstorm calculation, registry filters, and historical analysis.

Expected outcome: pass/fail result must be recorded with endpoint, UI location, dataset/source, and reviewer-visible evidence.

File Upload Testing

Upload CSV/XLS/XLSX files through Analyze Historical Dataset. Confirm station detection, parameter detection, missing values, duplicate rows, quality score, registry generation, and latestFileAnalyzed update.

Expected outcome: pass/fail result must be recorded with endpoint, UI location, dataset/source, and reviewer-visible evidence.

Forecast Simulator Testing

Run the simulator for Visakhapatnam and Machilipatnam. Confirm deterministic outputs, threshold trace, probability explanation, forecast result, and verification result.

Expected outcome: pass/fail result must be recorded with endpoint, UI location, dataset/source, and reviewer-visible evidence.

Verification Testing

Run threshold research by station/season and verify CSI, HSS, POD, FAR, and BIAS change deterministically with threshold input.

Expected outcome: pass/fail result must be recorded with endpoint, UI location, dataset/source, and reviewer-visible evidence.

CAPE Traceability Testing

Call ``/cwc/cape-traceability``, check NOW through T-6 timeline, source status, cache age, delta CAPE, and `STATIC_DATA_WARNING` behavior.

Expected outcome: pass/fail result must be recorded with endpoint, UI location, dataset/source, and reviewer-visible evidence.

SPA Deployment Testing

Open Render URL at ``/``, ``/dashboard``, ``/research``, ``/archive``, and ``/forecast``. Refresh each route and confirm React shell loads.

Expected outcome: pass/fail result must be recorded with endpoint, UI location, dataset/source, and reviewer-visible evidence.

API Preservation Testing

Call ``/system-status``, ``/history``, ``/forecast``, ``/cwc/historical-dates``, and ``/cwc/historical-analysis``. Confirm JSON payloads remain active after static frontend serving.

Expected outcome: pass/fail result must be recorded with endpoint, UI location, dataset/source, and reviewer-visible evidence.

Websocket Testing

Connect to ``/stream/atmospheric`` and verify forecast/trend/escalation/cycle frames arrive or fall back cleanly.

Expected outcome: pass/fail result must be recorded with endpoint, UI location, dataset/source, and reviewer-visible evidence.

Reviewer Acceptance Testing

Enable IMD Review Mode and confirm the reviewer sees date, time, station, observed event, forecast result, verification result, meteorologist summary, and recommended action without diagnostics clutter.

Expected outcome: pass/fail result must be recorded with endpoint, UI location, dataset/source, and reviewer-visible evidence.

11. User Manual

StormSense AI User Manual

Audience: IMD/CWC reviewer, operational forecaster, academic evaluator, and demonstration presenter.

Login

1. Open the single Render URL.
2. Enter reviewer/operator credentials.
3. Confirm the operational HUD shows backend and websocket status.
4. Use guest/local state only for non-production demonstration when authorized.

Run Forecast

1. Open AI Prediction Engine or Thunderstorm Forecast Simulator.
2. Confirm station, cycle, and data source metadata.
3. Review CAPE, LI, PWAT, SWEAT, K-index, forecast result, probability, and recommended action.
4. Use Forecast Generated time and cycle lock to verify operational timing.

Analyze Historical Case

1. Open Research & Insights.
2. Select Historical Thunderstorm Archive.
3. Filter by date, station, season, and event type.
4. Open historical analysis to compare forecast reproduction with observed event.

Upload Dataset

1. Open Historical Thunderstorm Archive or Thunderstorm Forecast Simulator.
2. Use Analyze Historical Dataset.
3. Upload CSV, XLS, XLSX, radiosonde, or historical sounding file.
4. Review dataset quality summary, detected columns, registry, verification, and export options.

Review Verification

1. Open Forecast Verification.
2. Select station, season, thresholds, or historical case.
3. Review CSI, HSS, POD, FAR, and BIAS.
4. Use the results to decide whether thresholds require calibration.

Generate Bulletin

1. Open Auto IMD Bulletin Generator.
2. Review current station hazards and district impact logic.
3. Generate thunderstorm, lightning, heavy rainfall, coastal squall, or district summary outputs.
4. Export or copy the bulletin for reviewer discussion.

Export Reports

1. Use CSV, JSON, XLSX, PDF, or analysis export controls where available.
2. Confirm source dataset and station metadata.
3. Attach reviewer docket metadata when required.

Use Review Mode

1. Toggle IMD Review Mode from the header.
2. Confirm charts and diagnostics are hidden.
3. Review date, time, station, observed event, forecast result, verification result, meteorologist summary, and recommended action.

12. Developer Manual

StormSense AI Developer Manual

Project Structure

Area	Path	Responsibility
Frontend entry	frontend/src/App.jsx	Authentication, global telemetry state, websocket connection, and workstation module routing.
Sidebar	frontend/src/components/layout/Sidebar.jsx	Primary navigation module list.
Research center	frontend/src/components/modules/ResearchHub.jsx	Start Here, About, archive, simulator, verification, dataset explorer, reviewer dashboard, index guide, terminology.
Phase 3 ops modules	frontend/src/components/modules/Phase3OpsModule.jsx	Atmospheric intelligence, watchdesk, lab, bulletin, analog, Andhra monitoring, verification.
Backend app	backend/main.py	FastAPI routes, authentication, websocket, forecasting, archive endpoints, file upload endpoints, static SPA serving.
Research engines	backend/analysis_engines.py	Historical observations, verification, probability, climatology, analog, district impact, bulletins.
Thermodynamics	backend/thermo.py	Sounding parsing and thermodynamic solver.
Sounding ingestion	backend/fetch_sounding.py	Wyoming fetch, cache metadata, freshness scoring, cycle-aware cache.
Database layer	backend/database.py and backend/connection_pool.py	SQLAlchemy PostgreSQL connection and Supabase schema initialization helpers.

Local Development

1. Install backend requirements from ``backend/requirements.txt``.
2. Install frontend dependencies from ``frontend/package.json``.
3. Set ``.env`` values for Supabase/PostgreSQL and authentication secrets.
4. Run FastAPI with ``uvicorn backend.main:app`` from the repository root.
5. Run ``npm run dev`` in ``frontend/`` only for local frontend development; production uses FastAPI static serving.

Code Organization Rules

- Keep scientific functions in ``analysis_engines.py``, ``thermo.py``, and ``fetch_sounding.py`` deterministic.
- Keep UI refinements inside existing components unless a component already owns the workflow.
- Do not hardcode deployment URLs in frontend code; production resolves same-origin through ``environment.js``.
- Preserve package-safe backend imports because Render starts ``uvicorn backend.main:app``.

Database Migration

- Supabase schema is maintained in ``supabase_schema.sql``.
- Migration execution is handled by ``migration_runner.py``.
- Runtime database access uses SQLAlchemy through ``backend/database.py`` and ``backend/connection_pool.py``.

Contribution Guidelines

- Run frontend build before deployment handoff.
- Run backend import and compile validation after backend changes.
- Preserve CAPE, probability, verification, and archive determinism.
- Add documentation evidence for any new operational module or endpoint.

13. IMD Demonstration Guide

Demo Scenario 1: Historical Event Verification

Open Research & Insights -> Historical Thunderstorm Archive, select a 2025 or 2024 event, inspect observed event, forecast reproduction, verification result, and threshold trace. Talking point: archive records participate in latest-event logic and thunderstorm registry filtering.

Demo Scenario 2: Forecast Simulation

Open Thunderstorm Forecast Simulator, select station and inputs, run simulation, and explain CAPE/LI/PWAT/SWEAT/K-index influence on forecast classification and recommended action.

Demo Scenario 3: Dataset Upload Analysis

Upload a CSV/XLS/XLSX historical dataset through Analyze Historical Dataset. Review records, missing values, duplicates, quality score, stations detected, date range, registry, and verification.

Demo Scenario 4: Operational Monitoring

Open Coastal Thunderstorm Monitoring Center or Live Operational Nowcast Center. Demonstrate Visakhapatnam/Machilipatnam focus, marine inflow, convergence, lightning/rainfall corridors, district impacts, and squall propagation guidance.

Expected Outputs

- Forecast classification and probability.
- Meteorologist explanation and recommended action.
- Verification result and skill metrics.
- Thunderstorm registry records.
- Exportable reports and bulletins.

Reviewer Questions

Question	Suggested Evidence
Where does CAPE come from?	CAPE traceability endpoint, sounding metadata, cache/source status.
How are forecasts verified?	Contingency matrix, CSI/POD/FAR/HSS/BIAS, historical analysis workflow.
Can uploaded datasets be reviewed?	File Analysis Center and <code>`cwc/analyze-historical-dataset`</code> .
How does deployment work?	<code>`render.yaml`</code> , single FastAPI service, static SPA fallback.
Does the system preserve IMD review usability?	Review Mode, Reviewer Dashboard, Start Here, About page.

14. Appendix: Project Inventory

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